Greyhound Blood Bank LIS

Clinical User Guide

Hospital transfusion-service laboratory information system

Compliance-supporting transfusion-service software. Not FDA-cleared, AABB-accredited, or CAP-certified by publication of this guide.

This guide describes the screens in the application. Facility procedures, medical-director policy, and the validation record still govern clinical use. Where a behavior depends on a local decision, the topic says so.

Figures show the development training database. They are not patient records from a clinical service.

Contents

Getting started

About this guide 3
Signing in 4
Dashboard 5
Using help 6

Patients and orders

Finding and registering patients 7
The patient chart 8
Visits 9
Orders 10
Specimens 11
Tests on the chart 12
Test history 13
DAT and elution 14
Products on the chart 15
Product history 17

Testing

Test worklist 17
Antibody identification worklist 19
Antibody identification 20
Unit antigen typing 21

Inventory

Inventory 22
Deliveries 23
Enter units on a delivery 24
ISBT receive 25
The unit chart 26
Modify products 27
Reagent QC 29
Stock par 29

Compatibility and issue

Compatibility testing 29
Issue and transfuse 30

Quality and operations

Transfusion reactions 32
Lookback and traceback 34
Deviations 35

Interfaces and reports

Printing 36
Billing review 37
Reports 38
Running a report 38
HL7 interface console 39
Instrument interfaces 40
Instrument simulator 41

Supervision

Supervisor queue 41
Shift handoff 42
Audit trail 43
Downtime snapshot 43

Safety

Safety checks 45

Appendix

Training data 46
When something goes wrong 47

Getting started

About this guide

What Greyhound Blood Bank LIS is, who this guide is for, and what the software does not claim.

Greyhound Blood Bank LIS is a hospital transfusion-service laboratory information system. It supports patients, specimens, immunohematology, compatibility, inventory, issue, transfusion documentation, lookback, reactions, and quality deviations.

This Clinical User Guide is for people who do that work: technologists, supervisors, and staff who review

billing, interfaces, or the audit trail. Configuration of catalogs, rules, users, and interfaces is in the Administrator Guide.

The system is compliance-supporting. Publishing this guide does not make the software FDA-cleared, AABB-accredited, or CAP-certified. Your facility's procedures, medical director, and validation record decide how you use it.

> Facility policy. Items that depend on a local decision are marked Facility policy. Do not treat those settings as universal rules.

The software is not a blood-collection establishment. It does not determine donor eligibility.

How to read a topic

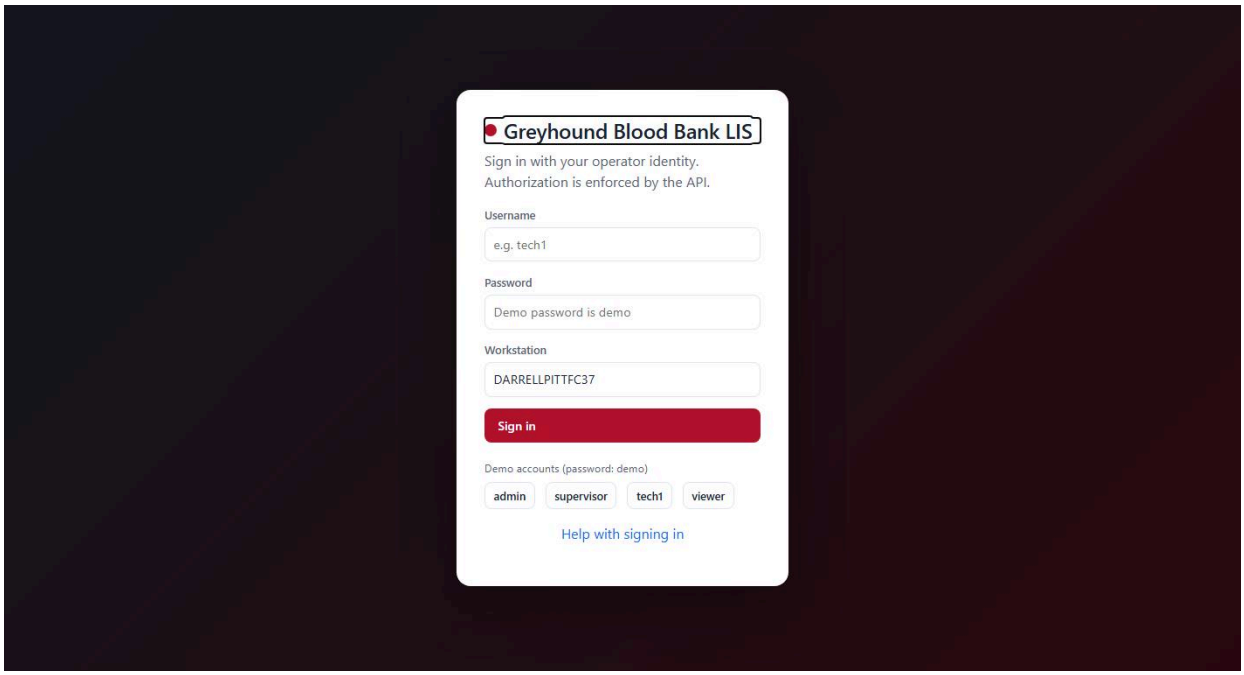
Each topic names the screen, what you do there, and what the system will block. **Hard stop** means the action cannot proceed. **Warning** means a supervisor can override it with a reason and, when required, an electronic signature. The screen explains the block in the alert. You do not need a code book to finish the task, but the safety topic lists the checks that come up most often.

Related guides

Open **Help** in the sidebar for both guides, search, and the PDF downloads. The PDFs contain the same topics as the screens.

Signing in

Sign in with your username, password, and workstation. Sessions expire, and five failed attempts lock the sign-in.



Open the application. If you are not signed in, the sign-in screen asks for three things.

1. **Username.** This is the account your administrator created. It is not your display name.
2. **Password.**
3. **Workstation.** This is recorded on the audit trail with your actions. The field starts as the computer name. Change it when you are working at a different bench or when your procedure names the workstation.

Choose **Sign in**. A wrong password or an unknown account shows an error and does not open a session. After five failed attempts the account is locked until an administrator unlocks it.

After you sign in

The sidebar shows only the areas your permissions allow. The dashboard lists those permission codes under **Your access**. A role change does not apply to a session that is already open. Sign out and sign in again after an administrator changes your roles.

How long a session lasts

The session ends after 30 minutes without activity, and it ends 12 hours after sign-in even if you are still working. Signing out, an administrator lock, or deactivating the account also ends it. When the session ends, the next action returns you to the sign-in screen. Finish or save the work you are in before you leave the bench.

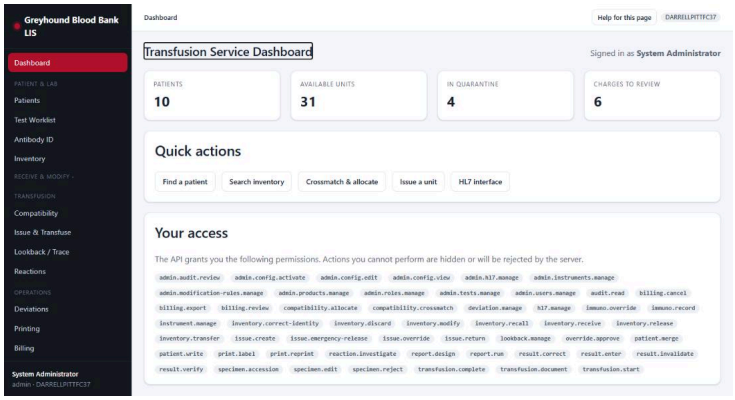
> Warning. Do not share a signed-in session. The audit trail records the signed-in user as the person who performed the action.

Training database

A training or development database may show demo accounts on the sign-in screen. Those accounts are described in Training data. Do not use a demo password on a clinical system.

Dashboard

The home screen shows queue counts, shortcuts into the day's work, and the permissions on your session.



The dashboard is the first screen after you sign in. It is titled **Transfusion Service Dashboard**.

What the counts mean

The cards summarize work that is waiting, including patients on file and units that are available or in quarantine, plus charges waiting for review when you can see billing. The numbers follow your permissions. A count you cannot act on is omitted rather than shown as a link you cannot open.

Use a card or the sidebar to open the queue. The sidebar is the full map: Patient and Lab, Transfusion, Operations, and Administration. **Help** is always present.

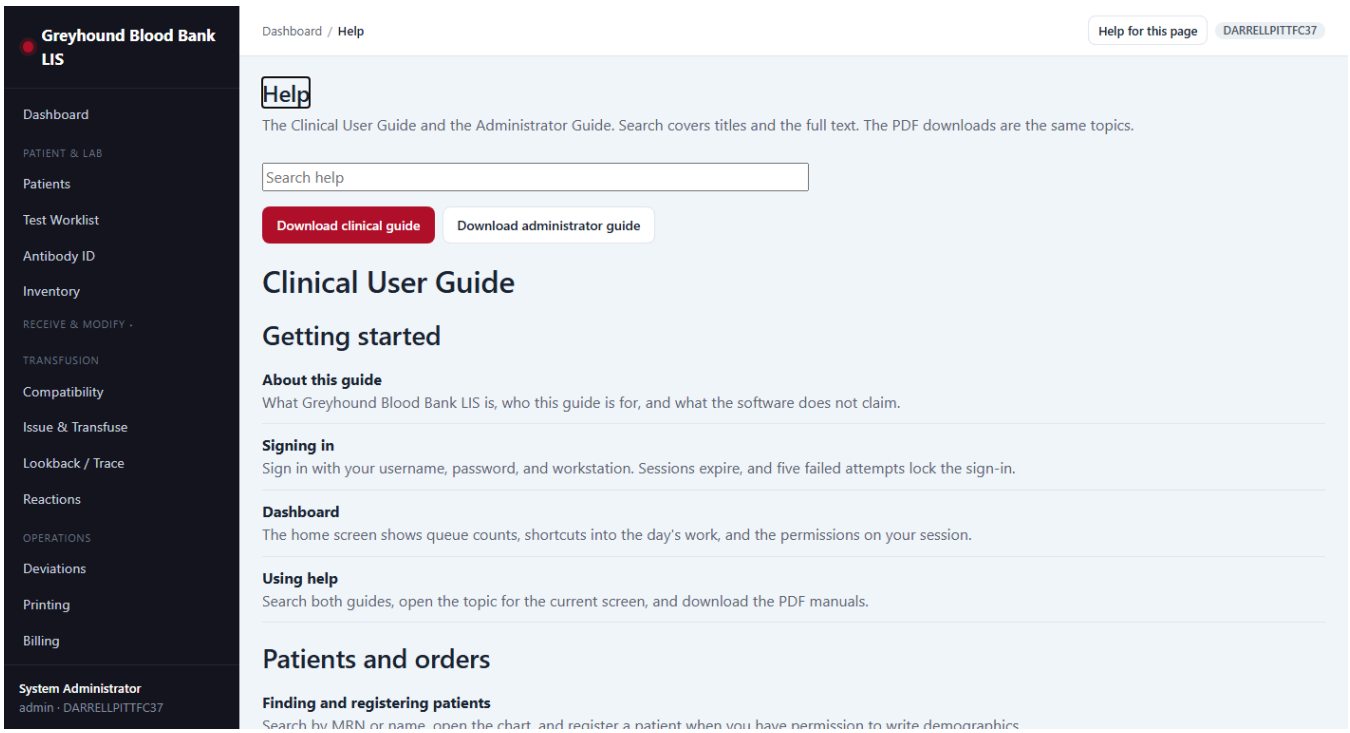
Your access

Your access lists every permission code on this session. The names are the codes the system checks, such as **result.enter** or **issue.create**. Roles are bundles of those codes. If a button on another screen is missing, compare this list with the task. An administrator changes what you can do on **Users** and **Roles**. You will not see Administration at all without an administration permission.

The top bar shows the workstation stored at sign-in and **Help for this page**, which opens the topic for the screen you are on.

Using help

Search both guides, open the topic for the current screen, and download the PDF manuals.



Help in the sidebar opens this library. You can also open it from the sign-in screen before you have a session, from the link **Help with signing in**.

Find a topic

Type in **Search help**. The search looks at the title, the short summary, and the body of every topic in both guides. Clear the box to see the full contents again. Topics are grouped the same way as the PDF table of contents.

Open a title to read the topic. **Open this screen** appears when the topic has a fixed address, such as Inventory or Facility Policy. Topics that depend on a patient or a unit do not offer that button. Use **Help for this page** while you are already on that chart or unit.

The previous and next links at the bottom follow the guide order.

Help for this page

On any signed-in screen, **Help for this page** in the top bar opens the topic that matches the address, including the patient-chart tab you have selected. Unit tabs and the inventory retype and quarantine boards do the same.

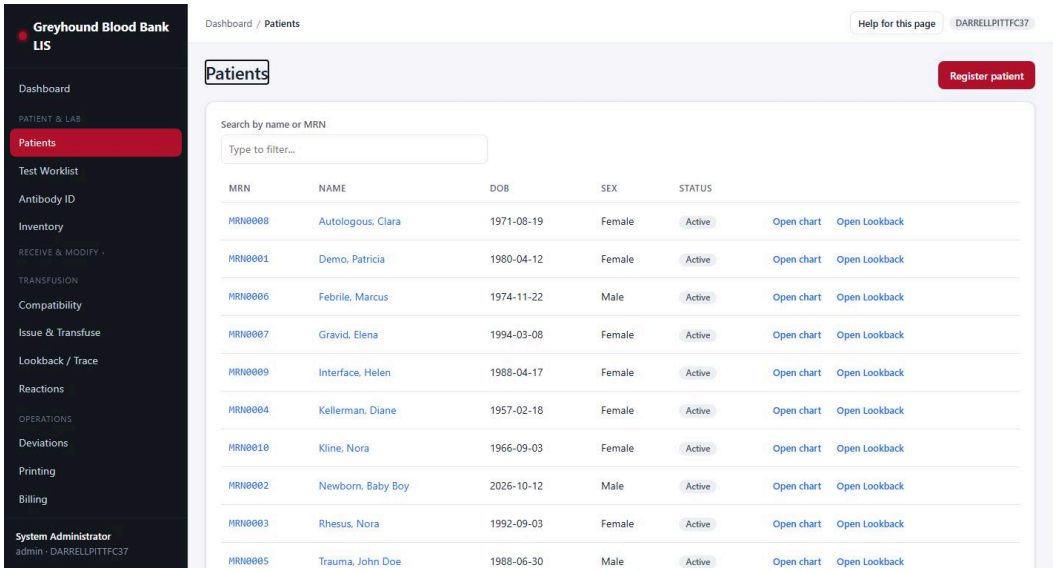
Downloads

Download clinical guide and **Download administrator guide** save the PDF manuals. They are generated from these topics, including the figures. Specialized authoring help on Order and Test Rules, Instrument Interfaces, and the SQL report designer stays on those screens and is linked from the administrator topics.

Patients and orders

Finding and registering patients

Search by MRN or name, open the chart, and register a patient when you have permission to write demographics.



Patients is the directory. Search with the medical record number or the name. Historical names remain searchable after a legal name change, so a former name can still find the chart.

Open a row to go to the chart. The chart is the place for visits, orders, specimens, results, and products. Do not create a second chart because the first search missed a former name.

Register a patient

Register patient is available when your session includes patient write permission. Enter the MRN, date of birth, last name, first name, and sex. Middle name and a comment are optional. Coded comments for the patient page can fill the comment from the catalog.

The MRN is the primary identifier you will type at issue and at transfusion documentation. Display names are not identifiers. Two patients must not share an MRN.

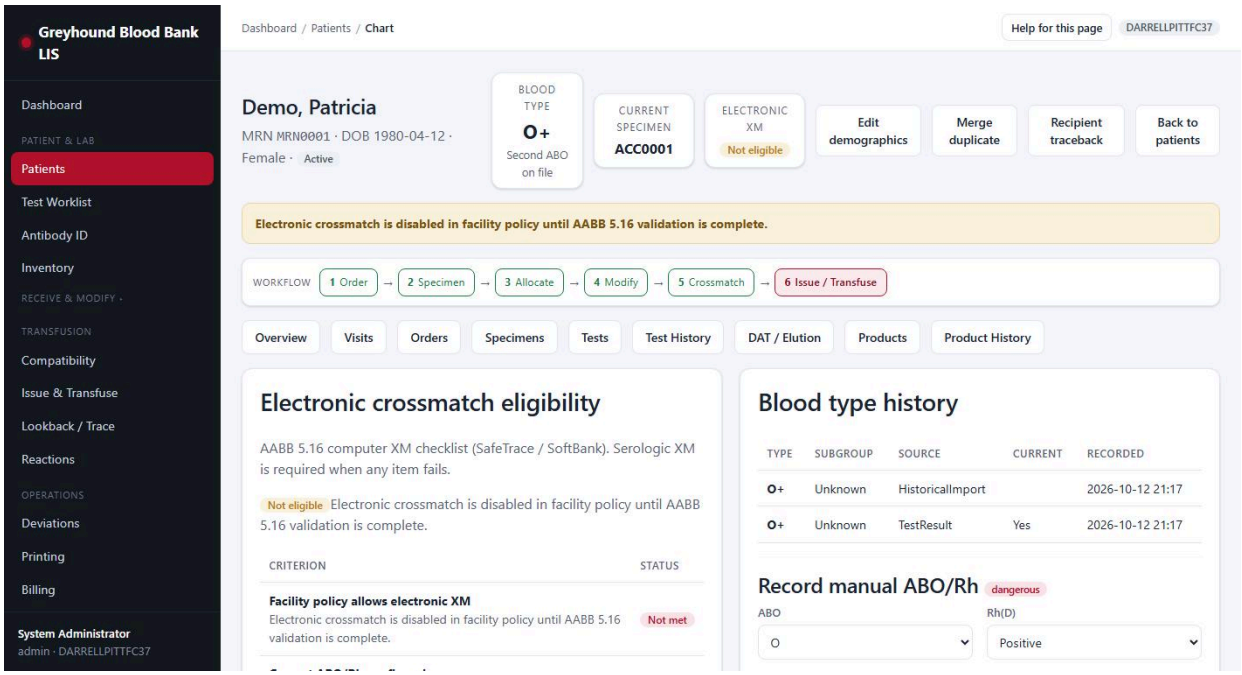
> Hard stop. You cannot issue, result, or document transfusion against the losing chart of a merged patient. Search brings you to the surviving chart. If the search still opens an inactive merged record, stop and use the surviving MRN.

What you cannot do here

Merging two charts is a supervisor action on the chart, not on this list. Correcting a name after registration is done on the chart and asks for a reason. See The patient chart.

The patient chart

The chart is the hub for identity, blood type, antibodies, special requirements, and every later step of the workup.



Open a patient from **Patients**. The overview is the default tab. The ribbon across the chart shows where this patient is in the transfusion workflow and jumps to the tab for that step.

What the overview shows

The header shows the current name, MRN, date of birth, and sex. Below that, the overview summarizes the facts the rest of the system will enforce:

- Current ABO and Rh, and whether history disagrees with it.
- Antibody history, including antibodies that are no longer detectable. A deactivated antibody still counts as history.
- Special requirements that are still active.
- Whether electronic crossmatch can be considered. Eligibility is calculated. It is not a checkbox you tick.

Use the tabs for the detail: Visits, Orders, Specimens, Tests, Test History, DAT / Elution, Products, and Product History. Each tab has its own topic in this guide.

Changing identity

Update demographics from the chart when your permission allows it. A name change asks for a reason. The previous name stays on the chart as an alias so old paperwork still finds the patient.

Merging is for a confirmed duplicate. It requires supervisor permission. You merge into the surviving chart. The losing MRN remains as a pointer and cannot be used for new work. Unmerge is available only when nothing has been recorded on the surviving chart because of the merge. If orders, results, or issues exist after the merge, do not unmerge. Correct the clinical record instead, and record a deviation if the merge was wrong.

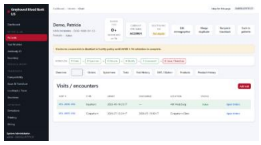
> Hard stop. Work entered on a merged-away MRN is rejected. Reprint labels and requisitions so the surviving MRN is what the bench and the ward scan.

Special requirements

Active special requirements (irradiated, antigen-negative, and the others your laboratory defined) are enforced when a unit is selected and issued. An end-dated requirement is inactive. Adding or ending one is audited. The catalog of requirement types is maintained in Administration.

Visits

Record the encounter the orders belong to, including location, attending provider, and pregnancy history when your procedure requires it.



The **Visits** tab lists encounters for this patient. An order is placed on a visit. Create the visit before you place the order when the patient has just arrived and no interface message has created one.

What to enter

Enter the visit identifiers your procedure uses, the ordering location, and the attending or ordering provider when you have them. Providers and locations come from the administrator catalogs. If the ward or the clinician is missing from the list, ask an administrator to add them. Do not invent a free-text location that will not match later issue documentation.

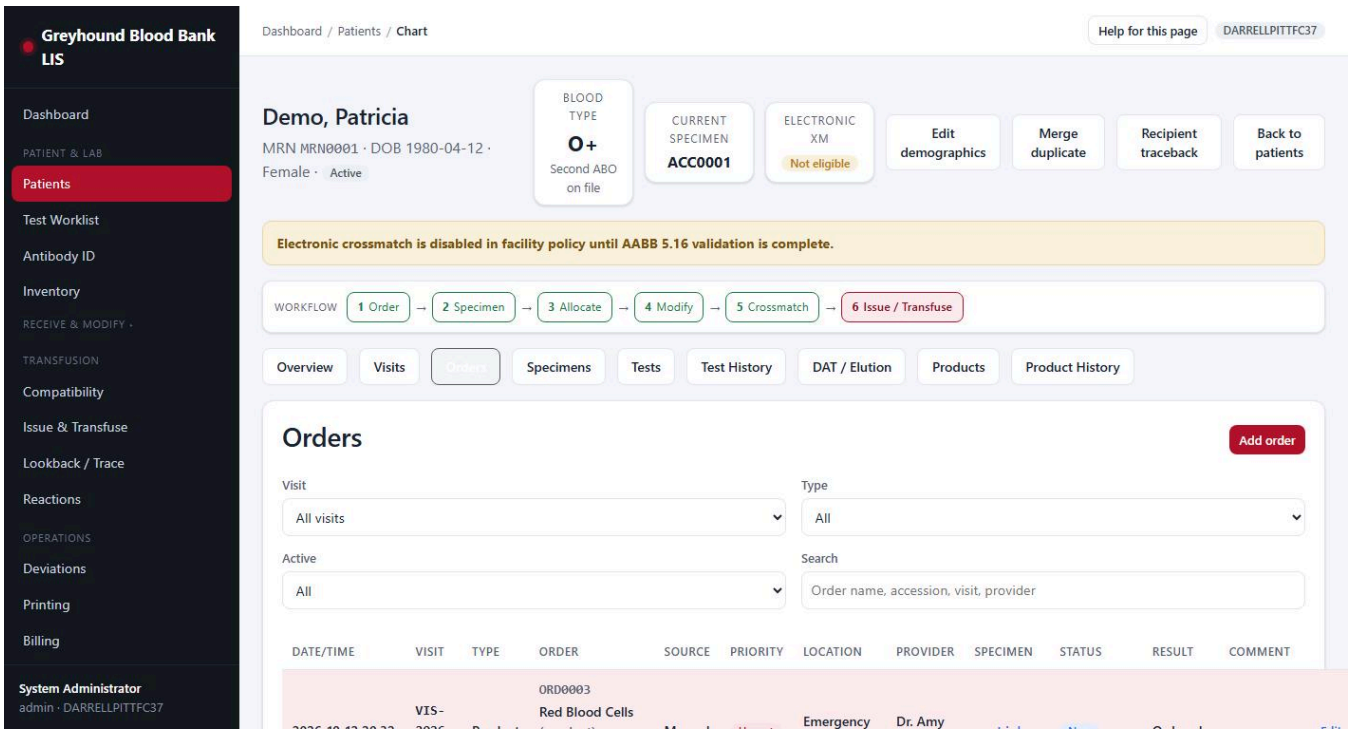
Pregnancy history, when the form shows it, is part of the clinical record used with Rh and antibody workups. Record what the history actually says. Do not leave a default that contradicts the interview.

Interface visits

An admission or registration message can create or update the visit before you open the chart. If the location or provider looks wrong, correct it here when you have permission. The correction is audited. The original interface message remains in the HL7 log for the interface analyst.

Orders

Place, update, and cancel orders, attach specimens, and add modification actions such as irradiate or thaw.



The **Orders** tab is where you place the work and see what is already ordered. Interface orders (ORM messages) appear here as well. You do not re-type an order that the interface has already filed unless your procedure says the electronic order is wrong.

Place an order

Select the visit, the ordering provider, and the test, test grouper, or product. A product choice keeps the product code. When the name is blank, the choice says Product and the id. The code is not shown as the name, and the stored product name is not changed. A grouper such as a type and screen expands into the member tests. Priority from an interface order, including STAT and ASAP, is stored on the order and shown on the worklist.

Product orders can carry modification actions: irradiate, wash, thaw, volume reduce, or leukoreduce. Those actions are defined in Administration. Completing the action later is done from inventory modification or the product workflow, and it can queue a charge when action billing is mapped.

Link a specimen

An order that needs a specimen stays incomplete until a specimen is linked. From the order you can open accessioning and collect the identifiers, collection time, and specimen type. See Specimens.

Hold and cancel

An order on hold does not accept bench work. Results cannot be verified while the order is on hold. Take the order off hold only when the reason for the hold is resolved.

Cancel requires a reason from the cancellation catalog. Cancellation does not delete the order. It records who cancelled it and why. Do not cancel an order to hide a verified result. Correct or invalidate the result instead.

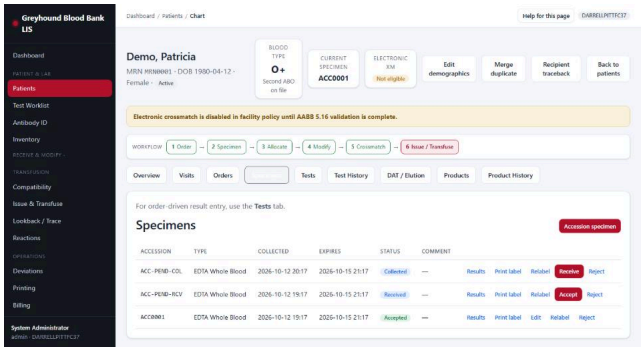
> Hard stop. Verification is blocked while the order is on hold, the specimen is rejected, or the specimen is past its testing window.

From the order to the unit

Product orders are what you allocate against on the Products tab and in Inventory. If the order is missing, allocation will not offer the patient. Place or release the order before you reserve a unit.

Specimens

Accession the specimen, track collection and receipt, and reject it when it cannot be tested.



The **Specimens** tab lists every specimen on the chart and lets you accession a new one when you are placing or completing an order. The pending-specimen queue on the test worklist is the bench view of specimens that still need to be received or accepted. Use whichever screen you are standing at. Both write the same record.

Accession

Record the specimen type, collection time, and the collector when your procedure requires it. The testing window is calculated from facility policy and the specimen type. You do not type an expiration that contradicts that rule.

Print the specimen label from **Printing** or the print action on the specimen. The label layout (what sits on the 2×1 inch label) is an administrator setting. You choose the printer and confirm the identifiers.

Receive and accept

A collected specimen is received into the laboratory, then accepted for testing. Until it is accepted, the worklist shows it as not ready. Reject the specimen when it is hemolyzed, mislabeled, or otherwise unsuitable. Rejection stops testing on that specimen.

> Hard stop. Results cannot be verified on a rejected specimen or on a specimen past its testing window. Collect a new specimen rather than overriding the window for routine work.

Relabel

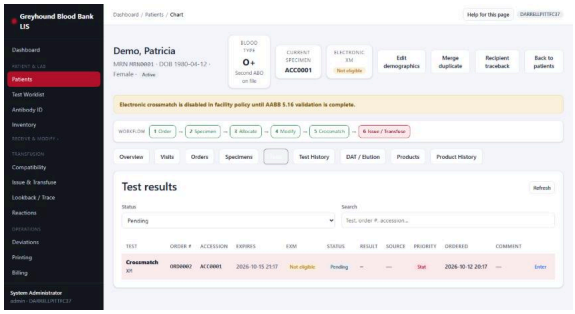
Relabeling changes the barcode. It requires a reason. Use it when the printed label is wrong or damaged, not to move a specimen onto a different patient. A specimen belongs to one patient. If it was accessioned to the wrong chart, reject it and accession a new specimen on the correct chart.

Identity at the bench

The accession number is what you will select when you record a crossmatch. At issue, the system checks that a current specimen exists for this patient and that it has not expired. A missing or expired specimen is not offered for routine issue.

Tests on the chart

See this patient's pending and completed tests, and open result entry without leaving the chart.



The **Tests** tab is this patient's view of the same work that appears on the Test worklist. Use it when you are already in the chart and need to see what is pending, what is blocked, and what has been verified.

Enter and verify from the row when your permissions allow it. The rules are the same as on the worklist:

- The person who entered a patient ABO or Rh result cannot verify that result.
- Verification is blocked on a rejected specimen, an expired specimen, or an order on hold.
- A delta between the new ABO or Rh and history stops verification until a supervisor records Retain or Replace and signs.

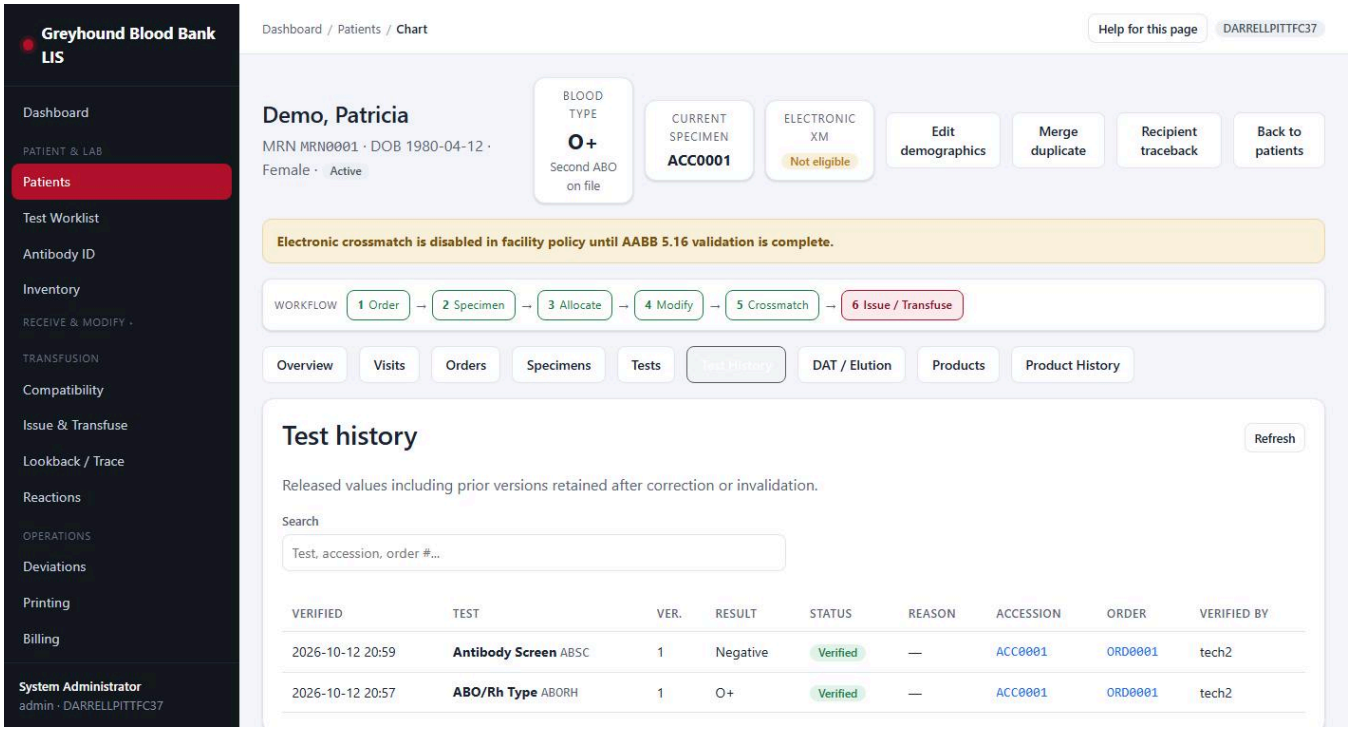
Correcting or invalidating a verified result is a separate action. Correction requires an electronic signature. Invalidation leaves the old result in history and marks it so later decisions do not treat it as current. Do not edit a verified value in place.

Interface results that are waiting for review show as such. Accepting an interface result is still verification. Read the transmitted values before you verify them.

The antibody chart is part of this patient's testing record. You can add an antibody or deactivate one when no antibody-identification workup is open. Deactivating an antibody does not erase it and does not make the patient eligible for electronic crossmatch.

Test history

Review verified, corrected, and invalidated results in the order they were recorded.



Test History is the longitudinal record. It shows what was verified, what was later corrected, and what was invalidated. Current blood type and antibody history on the overview are derived from this record. They are not a separate notepad.

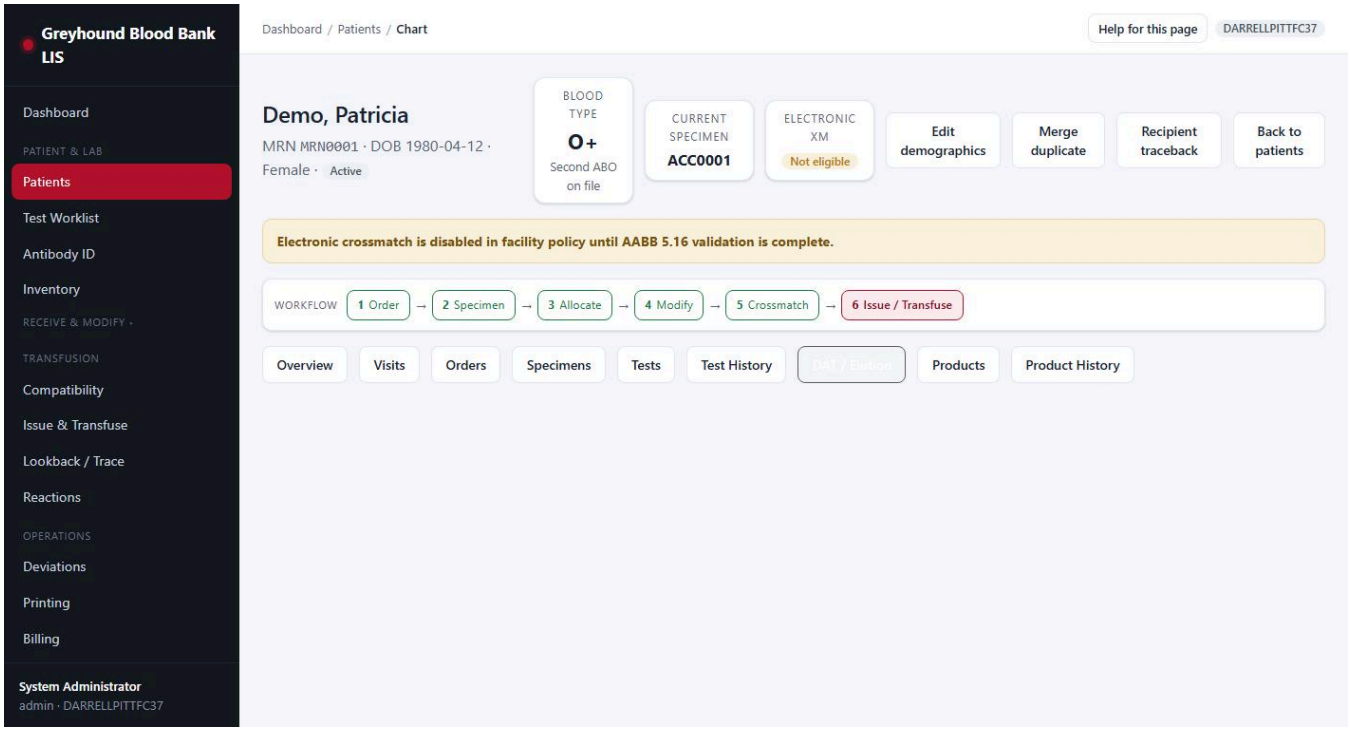
Use this tab when a result on the overview surprises you, when you are investigating a discrepancy, or when a clinician asks what the previous type and screen showed.

Each version keeps who entered it, who verified it, and when. A correction does not remove the earlier version. If you need to explain a change, open the version rather than relying on memory.

> Warning. An ABO or Rh discrepancy between the current result and history blocks routine issue until it is resolved with Retain or Replace. Test History is where you see both values.

DAT and elution

Record direct antiglobulin testing and elution work on the chart, separate from the antibody-identification workup.



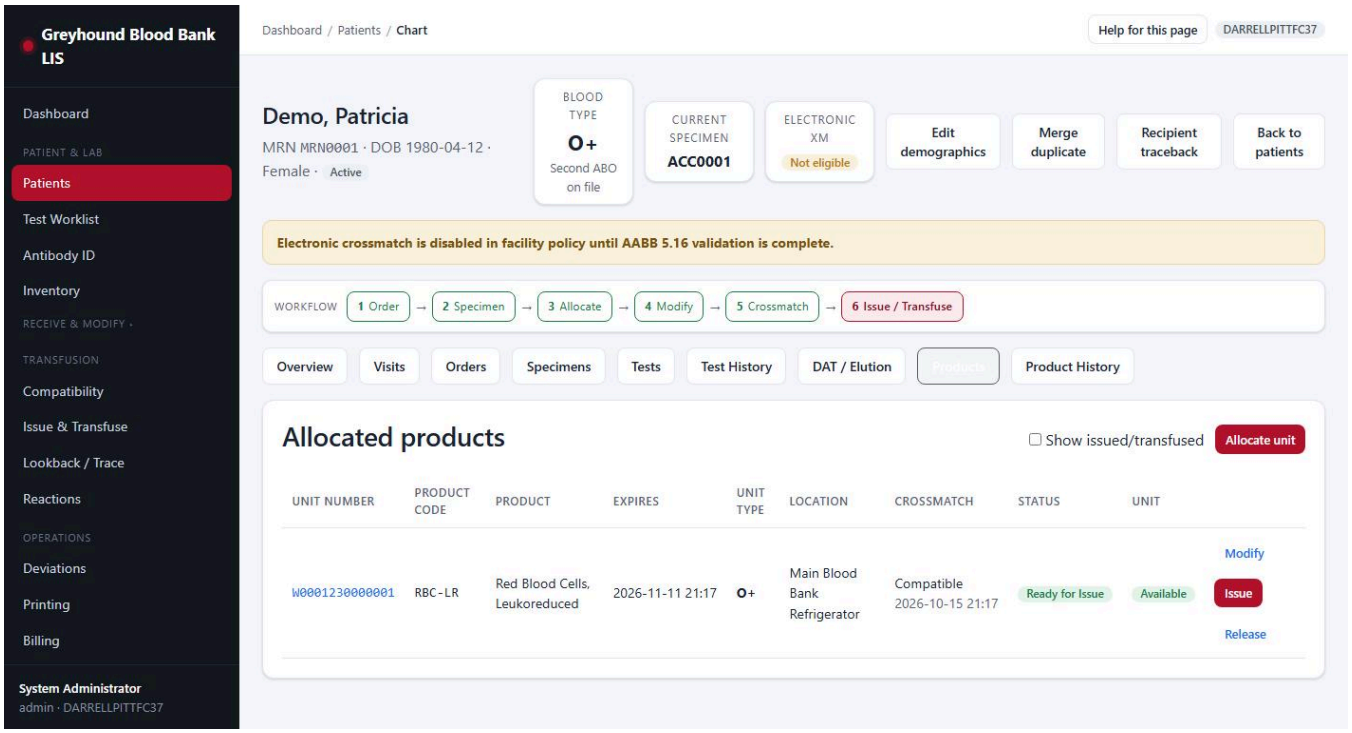
The **DAT / Elution** tab holds direct antiglobulin test and elution records for this patient. Use it when the workup is a DAT or an elution rather than a panel identification. Panel identification has its own workup under Antibody identification.

Enter the specimen, the method your procedure names, and the reactions you observed. Verify according to the same result rules as other blood bank tests: the specimen must be acceptable, the order must not be on hold, and you cannot verify your own patient ABO or Rh when that rule applies to the test.

A positive DAT does not by itself add an antibody to the chart. Identification of a specificity is done through the antibody-identification workup, which posts identified antibodies only when the workup is completed and reviewed. Do not type a specificity into a comment and treat that comment as antibody history.

Products on the chart

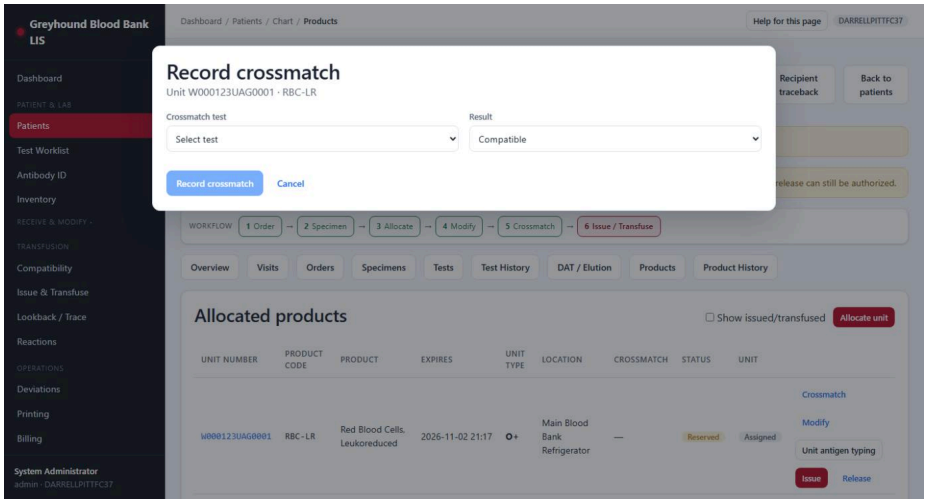
Reserve, crossmatch, and issue units from the patient's product list, including the issue and crossmatch dialogs.



The **Products** tab lists units reserved or issued to this patient. It is the chart-side view of allocation. The laboratory-wide queues live on Compatibility and Issue and transfuse.

What each row shows

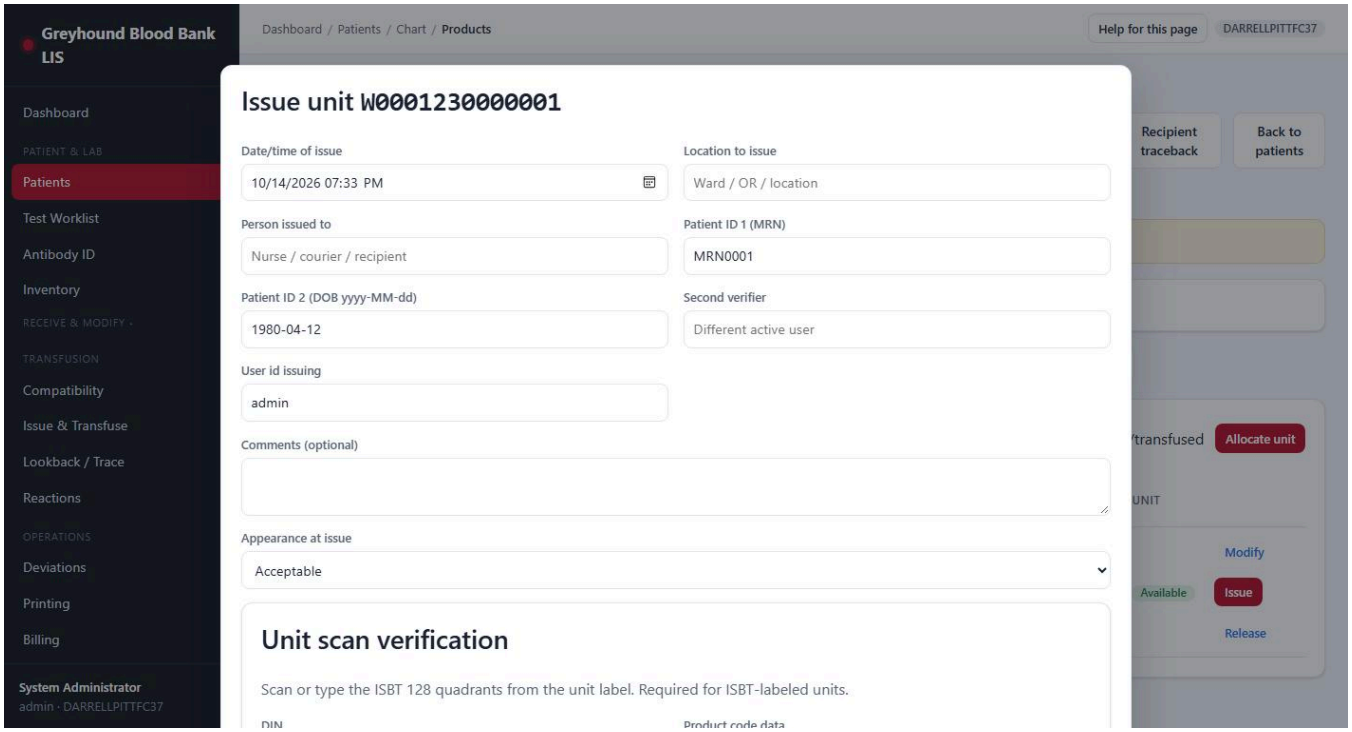
Each row shows the unit number, product, expiration, unit ABO and Rh, storage location, crossmatch result, and status. A row without a crossmatch result says so. A reserved unit that still needs a crossmatch offers **Crossmatch** when you have permission to record one.



The crossmatch dialog asks for the specimen and the method. Electronic crossmatch is offered only when the patient is eligible and facility policy allows it. Otherwise record a serologic crossmatch and the reactions your test definition requires. Saving the crossmatch updates the row.

Issue from the row

Issue opens the issue panel for that reserved unit. The patient and the unit are already selected. You still complete identity, appearance, and the other issue checks. The issue gate runs here the same way it runs on the Issue and Transfuse desk.



Enter two independent identifiers. MRN, date of birth, and name are the tokens the gate compares. A single checkbox is not an identifier. Record appearance. If the unit is going out in a cooler, record the cooler or transport id and the destination. When facility policy requires a second verifier or a scanned P-tag, those fields are mandatory before **Issue unit** is available.

If the gate returns a warning, the panel asks for an override reason and an electronic signature. A hard stop has no override. Read the message, fix the condition (specimen, type, allocation, or special requirement), and try again.

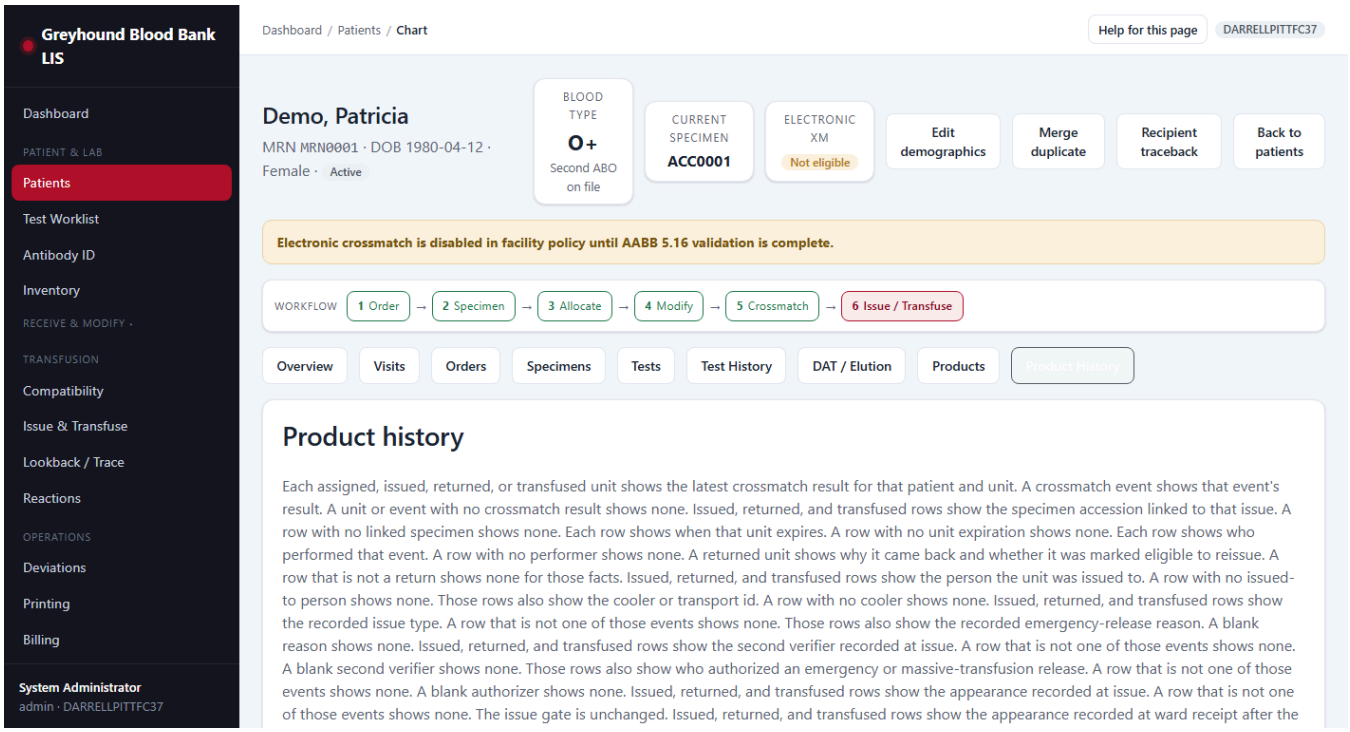
> Hard stop. Emergency release is not available from a routine row that the gate has blocked for a missing specimen, an outdated unit, or an incompatible allocation. Emergency and massive transfusion use the issue type described in Issue and transfuse, and they require the emergency-release permission.

After issue

The row shows the issue. When the unit is in transit, ward receipt is recorded on the Issue and Transfuse desk. Transfusion documentation can be started from a row that is issued and has been received when your procedure requires receipt first.

Product history

Review units already issued or transfused to this patient, including returns and final disposition.



Product History lists units that have already moved past the active reservation: issued, transfused, returned, or otherwise disposed. Use it when you are answering what this patient received, which unit numbers were involved, and whether a unit came back to inventory.

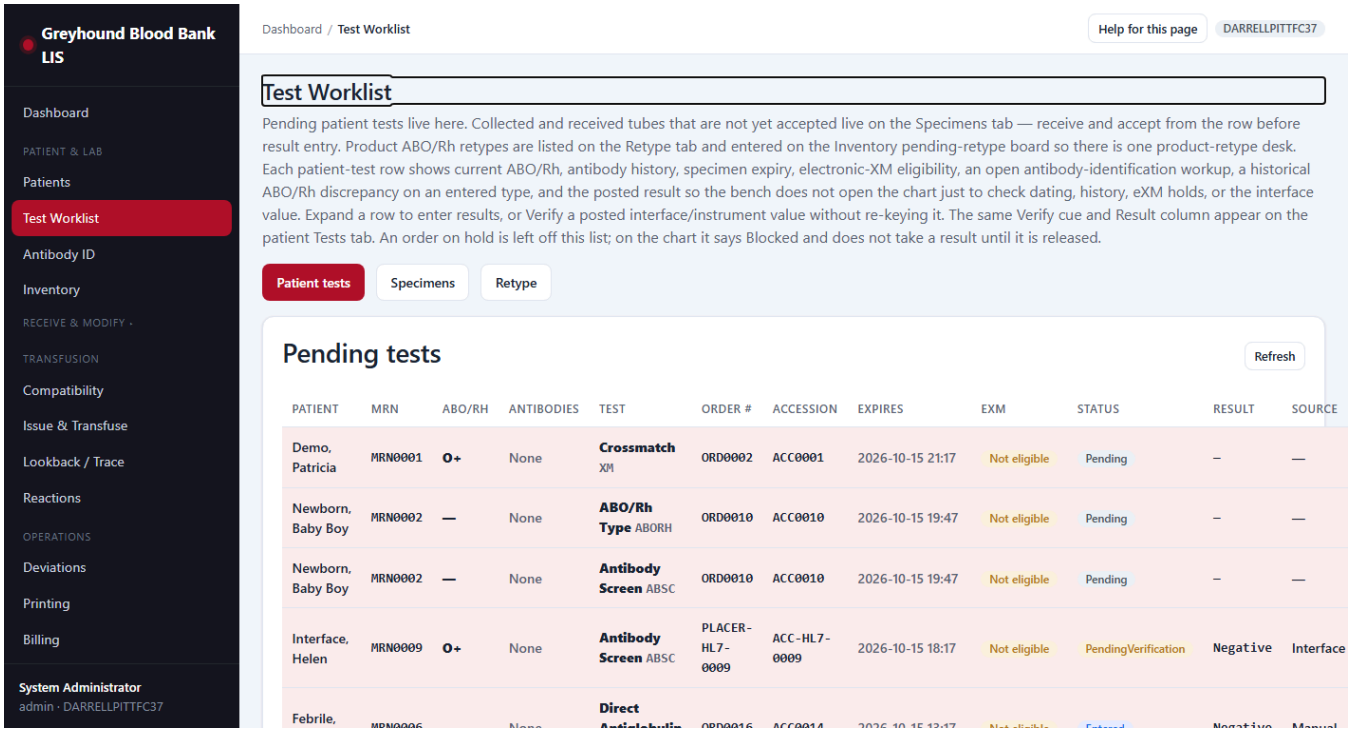
The active reservation list stays on the Products tab so the bench is not scrolling through old transfusions to find the unit in hand. Show issued and transfused on Products only when you turn that option on. History remains the place for the completed record.

Each line keeps the unit number, product, patient identifiers used at issue, and the disposition. This is also the record you will compare during a lookback or a reaction investigation. Do not delete a historical line. If something was documented in error, correct it with the appropriate return, reaction, or deviation workflow so the original entry stays visible.

Testing

Test worklist

Work pending specimens and results in priority order, and verify, correct, or hold them under the result rules.



The **Test Worklist** is the bench queue for specimens and results. It opens when you have permission to enter results. Rows are ordered so STAT and ASAP work is visible, and blocked rows stay visible with the reason rather than disappearing.

Specimens on the worklist

The specimen side lists collected specimens waiting to be received and received specimens waiting to be accepted. Receive, then accept, before you result. Reject from here when the specimen cannot be tested. The patient's chart updates with the same status.

Enter a result

Open the row, enter the values the test definition asks for, and save. Entry and verification are separate steps. You can enter a result you cannot verify. Another qualified person verifies it.

For patient ABO and Rh, the person who entered the result cannot verify it.

> Hard stop. Verify is refused when the specimen is rejected or expired, when the order is on hold, or when you are verifying your own patient ABO or Rh entry.

Discrepancies

If the ABO or Rh does not match history, verification stops. A supervisor chooses **Retain** (keep the historical type and do not replace it with this result) or **Replace** (accept the new type), with an electronic signature. Do not verify around the delta by editing history.

A manual ABO or Rh override, when policy allows one, is an immunohematology override. It also requires

the override permission and a signature. It is not the same action as ordinary verification.

Interface and retype rows

Results that arrived from an instrument wait for the same verification. The row marks them so you know you did not type them. Read the values.

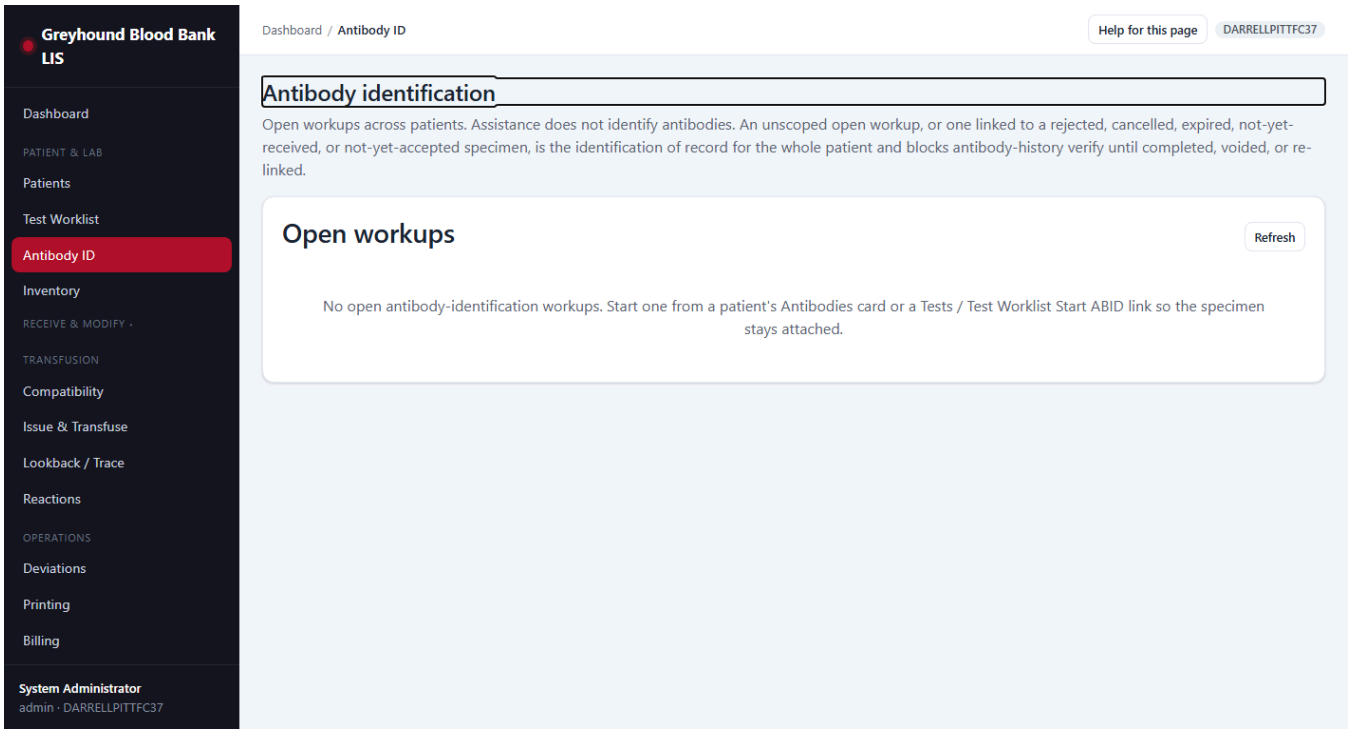
The product retype tab of the worklist lists units that still need a front type. Recording the retype is the same action as the retype board in Inventory. A match can release the unit. A mismatch quarantines it. An outdated unit that matches is expired rather than made available.

Correct or invalidate

After verification, a wrong result is corrected or invalidated. Correction requires a signature. Invalidation keeps the prior result visible and removes it from current history. Both are audited. The supervisor queue shows verifications that are waiting when your role is the one who verifies.

Antibody identification worklist

Find open antibody-identification workups and open the one you are going to result.



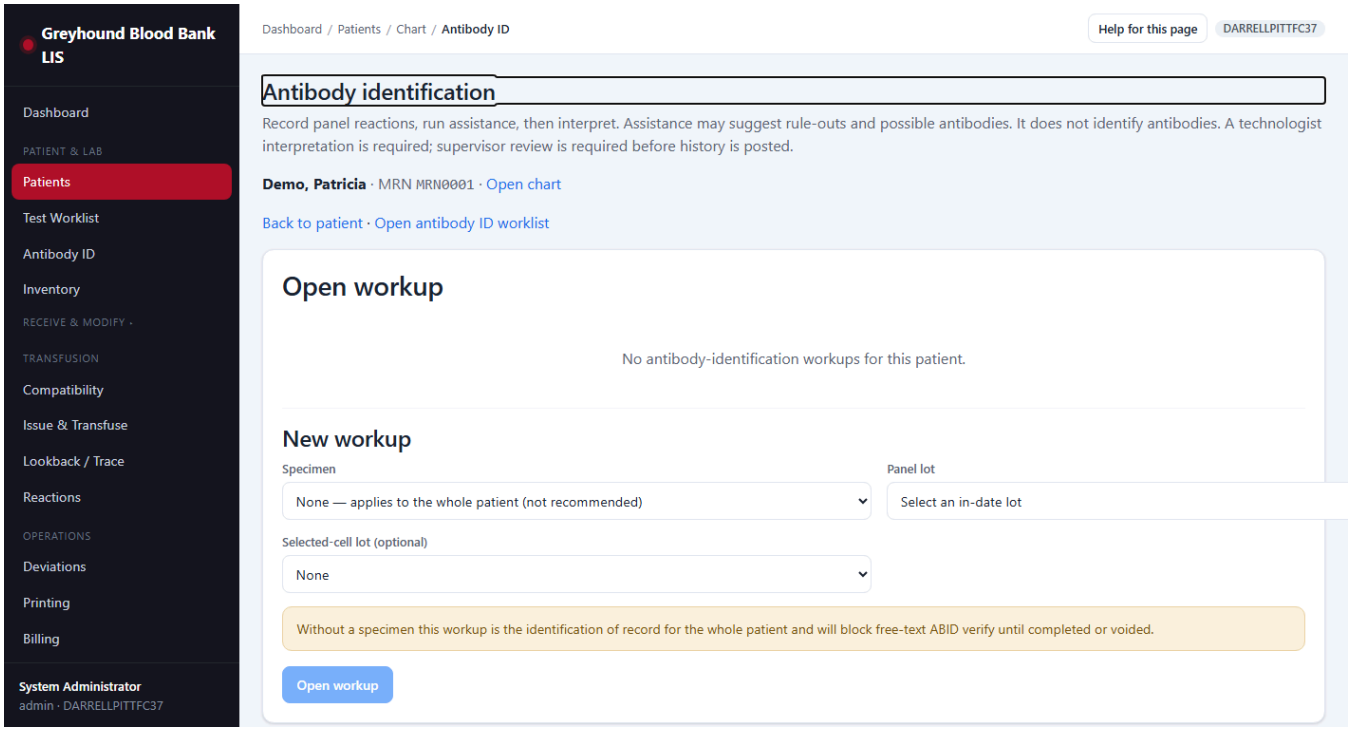
Antibody ID lists workups that are not finished. Filter to the patients or statuses you are responsible for. Open a row to go to that patient's workup.

You need permission to record immunohematology, or permission to override, to see this list. Recording reactions is not the same as identifying an antibody. The workup screen keeps those actions apart. See Antibody identification.

A patient with an open workup is marked on the issue desk. You can still issue, with that warning visible. You cannot add or deactivate chart antibodies while the workup is open, and you cannot record an electronic crossmatch for that patient until the workup is no longer open.

Antibody identification

Record panel reactions, use assist without letting it identify the antibody, and send the workup for supervisor review.



Open a workup from the worklist or from the patient chart. The screen shows the panel, the phases, and the reactions you enter. The lot must be an active antibody-panel lot. A withdrawn lot cannot be accepted or completed.

What you record

Select the specimen and the panel lot. Enter the graded reactions for each cell and phase. The phase list (immediate spin, 37°C, AHG, check cells, and any others your laboratory defined) comes from Administration.

Assist compares the pattern with the panel antigram and suggests specificities that fit. Assist does not identify the antibody. You still interpret. The identified specificity is posted to the patient's antibody history only when the workup is completed under your procedure, including supervisor review when policy requires it.

While the workup is open

You cannot add or remove antibodies on the chart. That prevents the history and the workup from

diverging. Serologic crossmatch can still be recorded. The issue desk shows an open-workup warning and still allows issue. Electronic crossmatch is a hard stop until the workup is no longer open.

> Hard stop. Electronic crossmatch is refused while an antibody-identification workup is open. Deactivating an antibody after the workup, or marking it currently undetectable, does not restore electronic-crossmatch eligibility.

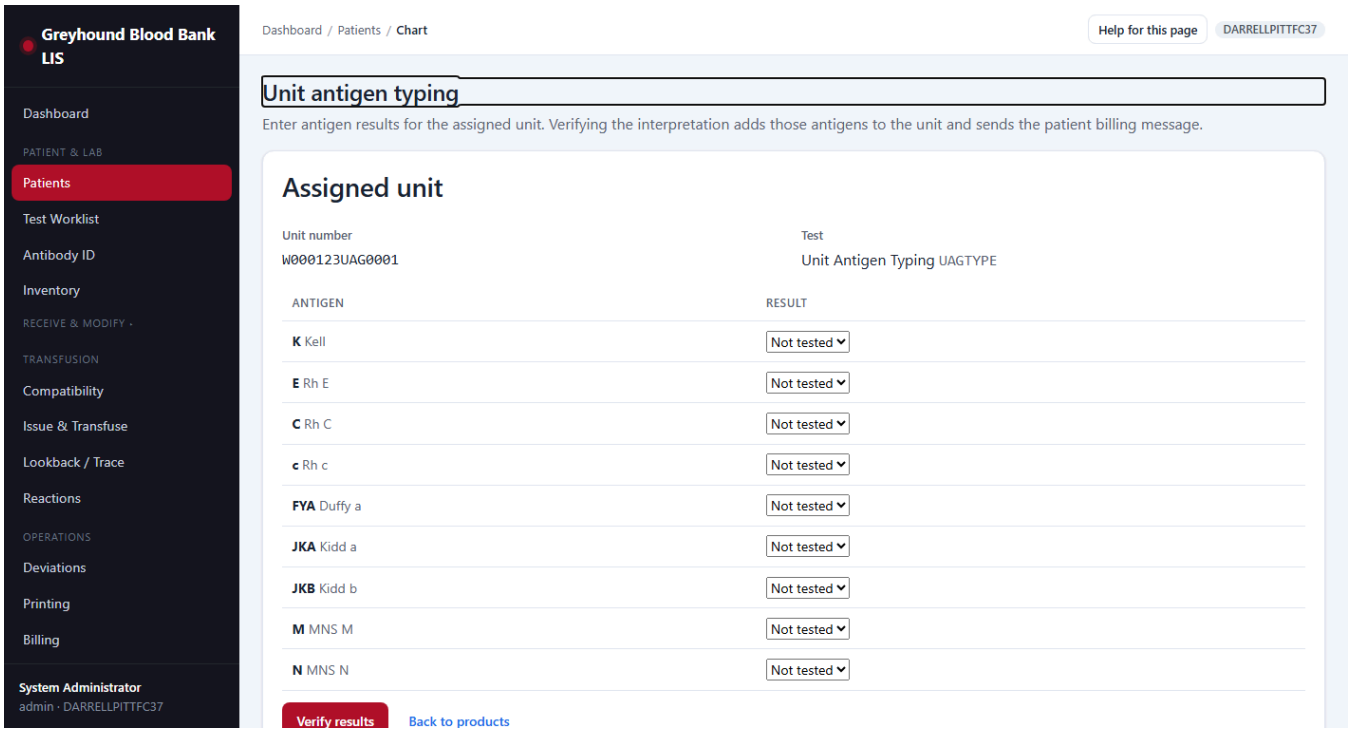
Supervisor review

When the workup needs review, it appears on the Supervisor queue. The reviewer confirms the interpretation. Completing the workup writes the identified specificities into antibody history as identified antibodies. Those rows drive antigen-negative selection and block electronic crossmatch from that point forward.

If the workup is closed without an identification, history is not given a new specificity. Say that in the workup comment so the next shift can see why the panel was closed.

Unit antigen typing

Record antigen typing performed on a specific unit for this patient.



Open unit antigen typing from the patient and the unit you are typing. This is a result on that unit, not a change to the patient's antibody history. Use it when you type a unit for an antigen the patient needs to avoid, or when you are confirming a labeled antigen.

Enter the antigen, the method, and the reaction your procedure uses. The unit's antigen record is what

allocation and issue compare with the patient's antigen-negative requirements. A unit that was not typed for a required antigen is not treated as antigen-negative.

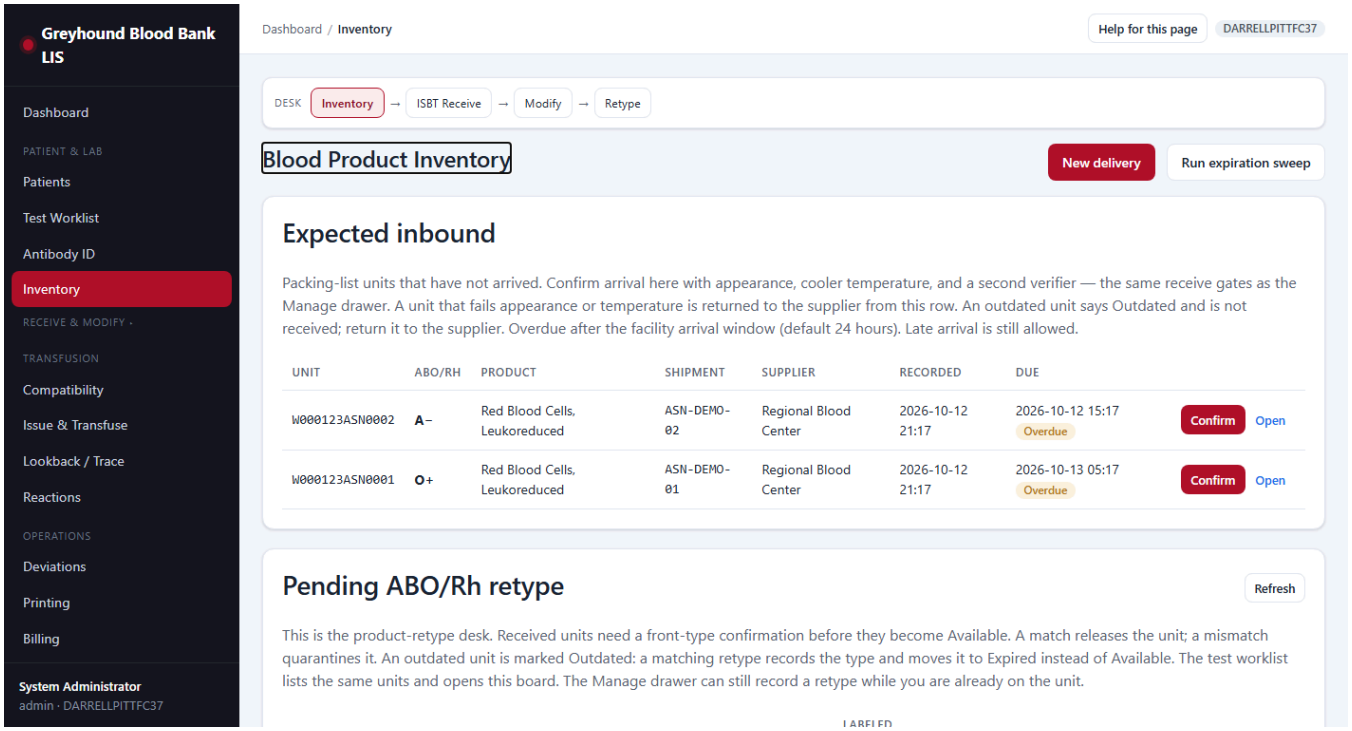
> Warning. An antigen-negative requirement that is not met is a warning at allocation and can block or warn at issue, depending on the rule. Typing the unit is what clears it. A comment on the chart does not.

The unit chart in Inventory shows the same attributes. You can open the unit from here to see location, status, and the rest of the unit record.

Inventory

Inventory

Search units, work the retype and quarantine boards, and open a unit or a delivery.



Inventory is the stock list. One board is open at a time: Search, Expected inbound, Retype, Near expiry, Quarantine, Discrepancy, Operational hold, and Open deliveries. Search opens first. Search by unit number, product, blood type, status, or location. Available units are the ones you can reserve. Quarantine, hold, missing, and damaged units stay out of routine selection until someone with release permission clears them.

The row shows the storage location by name. A missing name falls back to the location code so you can still see where the unit was filed.

Boards

Retype (**board=retype**, including a unit id) opens the front-type desk. Received units need a type confirmation before they become available. A match releases the unit. A mismatch quarantines it. An outdated unit that matches is marked expired instead of available. The test worklist lists the same units and opens this board.

Quarantine (**board=quarantine**) opens units held for a coded quality reason. The row shows product, location, volume, and any ABO or Rh retype mismatch. Release uses the same second-verifier gate as the unit's manage actions when policy requires two people. A unit created by a modification also asks for the unit number on the new label. That scan has to match the result. Other quarantine releases do not ask for it.

The other boards cover expected inbound units, units near expiry, units on hold, discrepancy units (missing or damaged), and open deliveries. Open the row rather than changing status from memory.

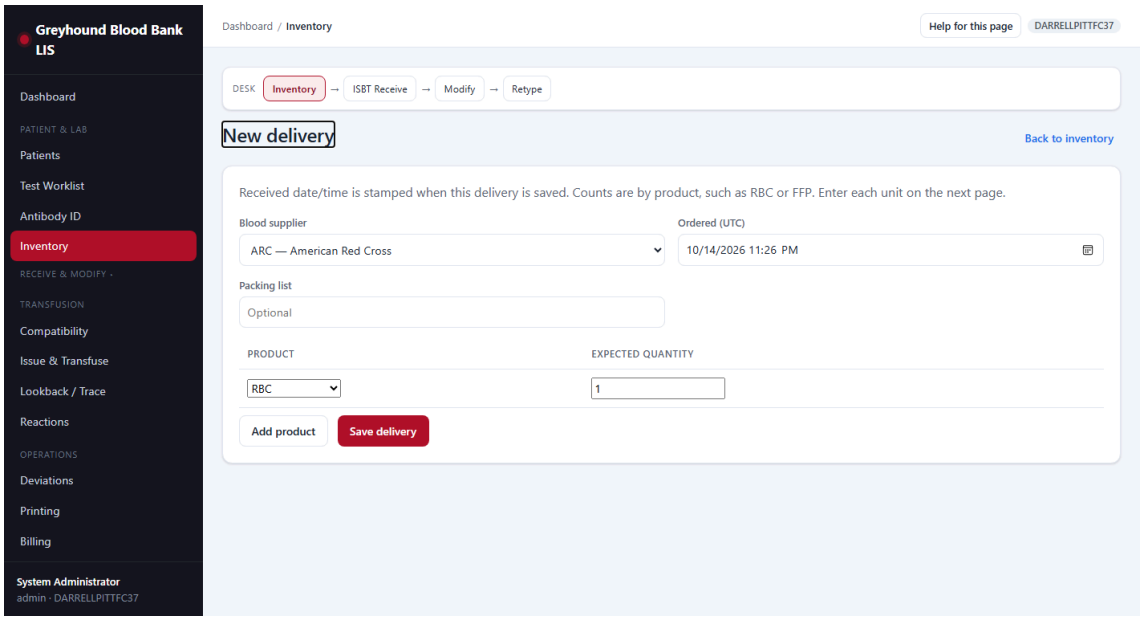
Start the next task

New delivery starts a shipment from a supplier. **ISBT Receive** and **Modify Products** are in the sidebar under Receive and modify. Open a unit number to see its full chart: status, attributes, history, and the actions that are legal for the state it is in.

> Facility policy. Which statuses can move to which other statuses is fixed. If an action is missing, the unit is not in a state that allows it. Do not expect a button for discard, transfer, or release on every row.

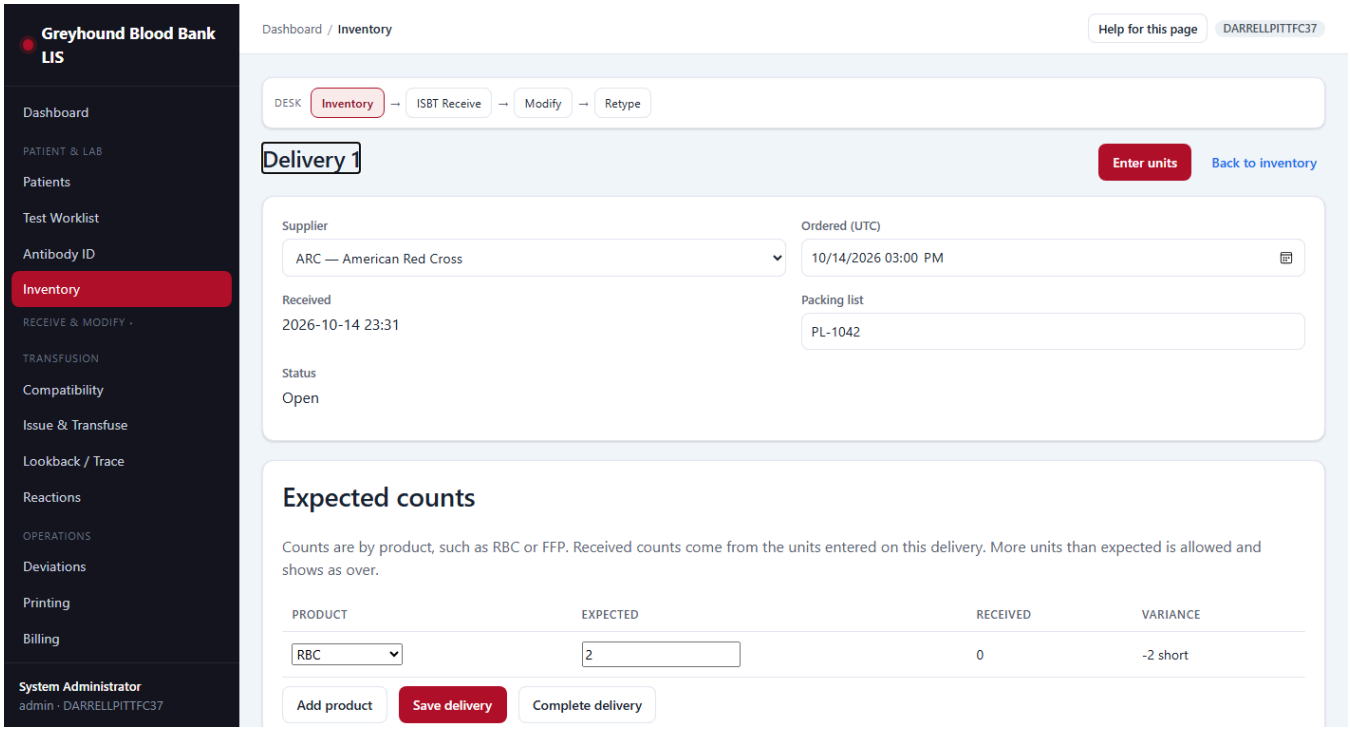
Deliveries

Create a delivery from a supplier and track the packing list before units are received into stock.



New delivery records a shipment you are expecting or have in hand. Choose the blood supplier. Suppliers are an administrator catalog. If the shipper is not listed, stop and have them added. Do not file the units under the wrong supplier to save the step.

The delivery page shows the packing list and the status of each expected unit. Units that were announced in advance (an advanced shipping notice) appear as expected inbound on the inventory screen as well.

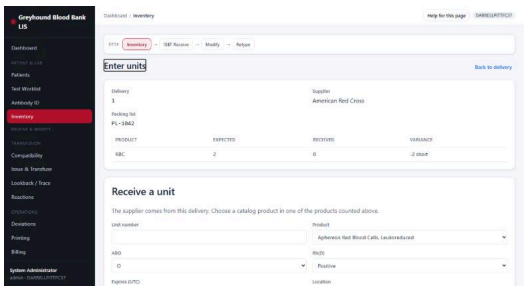


Open **Enter units** when you are ready to receive the physical units onto this delivery. Receiving checks temperature, appearance, and a second verifier when facility policy requires them. That work is on Enter units on a delivery. Scanning ISBT labels one unit at a time, outside a packing list, is ISBT receive.

Do not mark a delivery complete while units are still sitting on the counter unreceived. The expected-inbound queue is how the next shift knows what has not been booked in.

Enter units on a delivery

Receive each unit on the packing list, including visual inspection, temperature, and a second verifier when policy requires one.



This screen books the physical units on a delivery into inventory. Work one unit at a time. Confirm the unit number, product code, ABO and Rh, and expiration against the bag and the packing list. A mismatch is a reason to stop, not a reason to type what the paperwork says if the bag says something else.

Record appearance and temperature when the form asks. Facility policy can require a second person to verify the receipt. That person signs in as the second verifier. You cannot verify your own receipt when the policy requires two people.

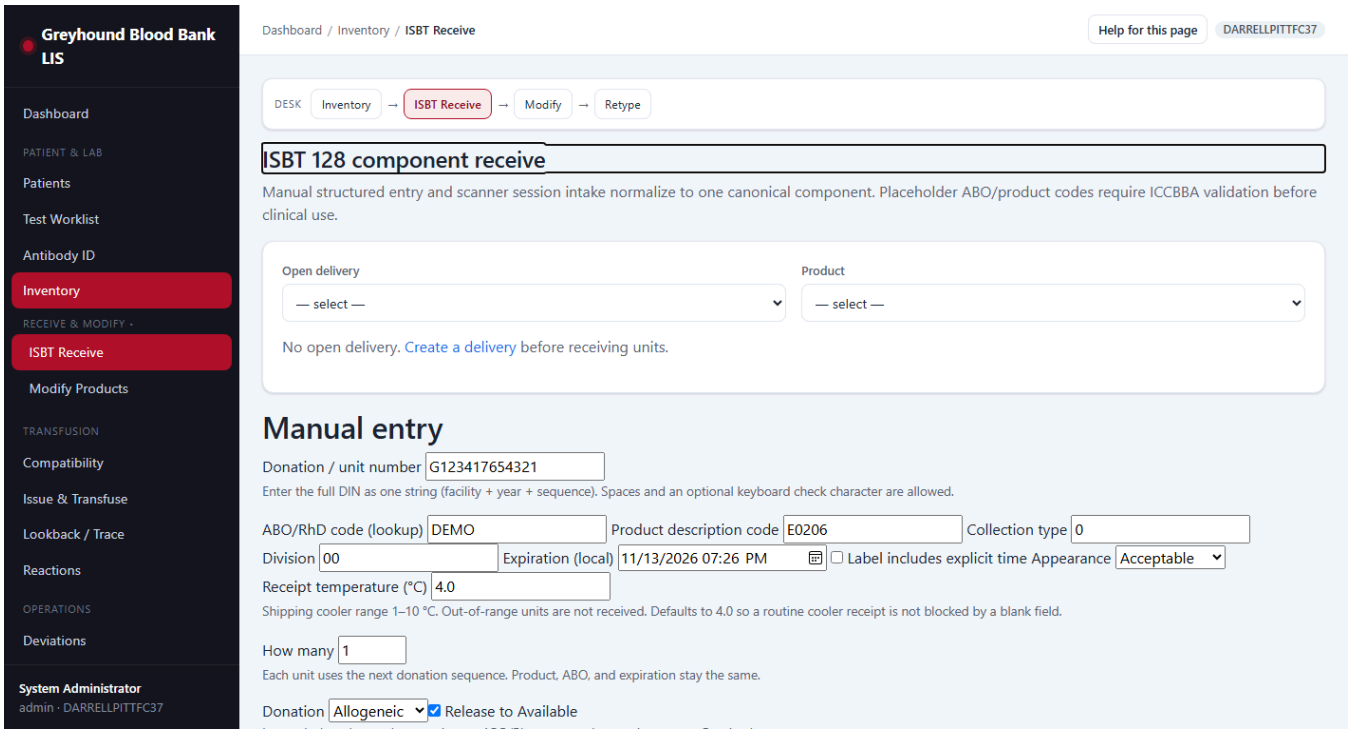
Received red cells and whole blood typically wait on the retype board before they are available. Plasma and platelets follow the release rule for that product. If the unit arrives damaged or out of temperature, quarantine or damage it here with the coded reason. Do not put it into available stock and hope someone notices.

Autologous and directed units must be tied to the intended patient at receipt. They will not issue to anyone else. Converting a directed unit to general allogeneic stock is a separate, permissioned action later, with a reason.

After receipt, open the unit chart if you need to print a component label or correct a data entry error. Identity correction (unit number or product code entered wrong) requires the inventory identity-correction permission and a reason. It is not an ordinary edit.

ISBT receive

Scan or type an ISBT 128 label to bring a unit into inventory without a packing-list delivery.



ISBT Receive is for scanner or keyboard entry of an ISBT 128 label: the donation identification number, the product code, and the expiration. Use it when the unit is not already on a delivery packing list. Use a delivery when you are working a shipment.

Start a scan session and scan the bag. The product code must exist in the ISBT catalog. The catalog shipped with the system is a placeholder until your facility imports a licensed ICCBBA extract. If a real product code is rejected as unknown, do not invent a local code that looks similar. Ask an administrator to import or add the code your license allows.

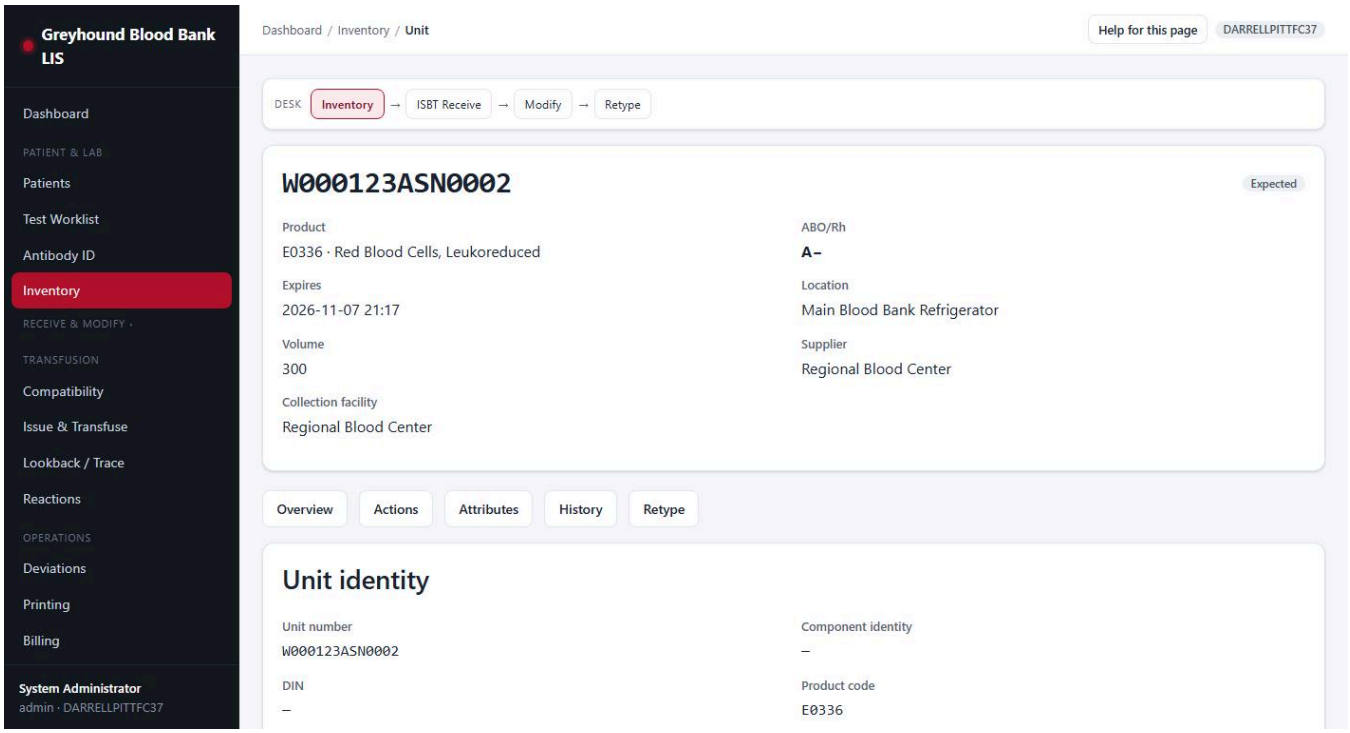
Confirm ABO, Rh, and expiration against the label after the scan. The scan can misread a damaged barcode. Keyboard entry is there for that case. Treat keyboard entry with the same second check you would give a scan.

> Facility policy. Official ISBT codes are licensed. The software does not include the ICCBBA tables. A code that is missing is a catalog gap, not a reason to relabel the unit.

Received units follow the same retype and quarantine rules as units received on a delivery.

The unit chart

Open one unit to see status, attributes, history, retype, and the actions that status allows.



Open a unit from Inventory, from a delivery, or from a patient's product row. The header shows the facility unit number, product, ABO and Rh, expiration, and status.

Tabs

Overview shows where the unit is, why it is quarantined or on hold if it is, and the receipt inspection (appearance, temperature, visual notes).

Actions are the moves that are legal from the current status: quarantine, hold, release, transfer to

another location, allocate to a patient order, discard, return to the supplier, or mark missing or damaged. Each action asks for the reason the rule requires. Discard and some releases need a higher permission than everyday receive and transfer. If the button is disabled, you either lack the permission or the unit status does not allow the move.

Attributes lists antigens and modifications (irradiated, leukoreduced, and the others recorded on the unit). Antigen typing you performed is recorded with Unit antigen typing or on the retype tab.

History is the status log. It is append-only. You cannot delete a prior status to tidy the story.

Retype records the front type. A mismatch quarantines the unit. Do not release a mismatch from this tab. Release, if your medical director allows it at all, is a quarantine-release on the supervisor queue with the second verifier.

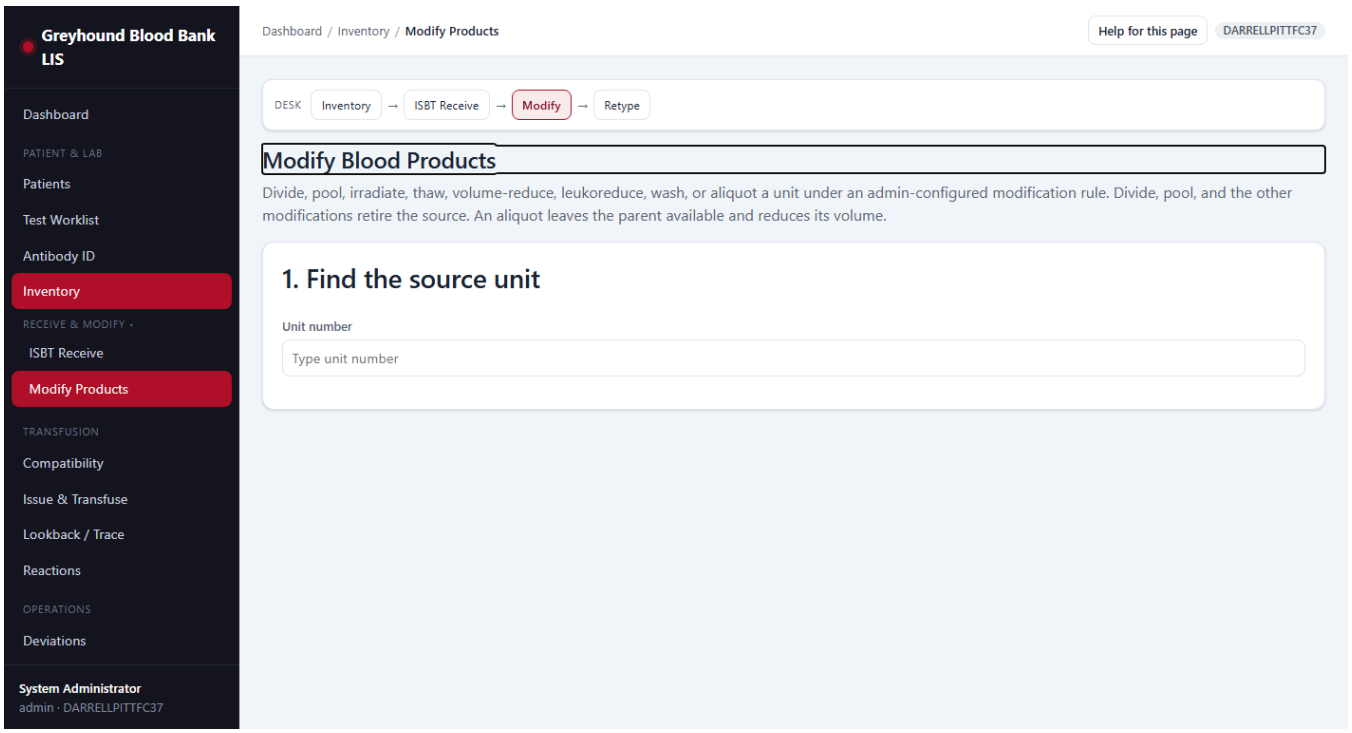
Allocate

Allocation from the unit asks for the patient order. The order must be an open product order for a patient who may receive this unit. Autologous and directed units only allocate to the intended patient. The screen warns when the unit is not typed for an antigen the patient must avoid, and when a red cell has not been retyped.

Printing a component label is available from the unit when you have print permission. Reprinting asks for a reason.

Modify products

Divide, pool, irradiate, thaw, wash, aliquot, volume-reduce, or leukoreduce a unit under the modification rules.



Modify Products performs component processing. Each modification is defined by an administrator rule: which source products are allowed, what the resulting product code is, and how the expiration changes. You pick the modification and the source unit. You do not invent a new product code at the bench.

What happens to the units

The source unit is consumed or reduced according to the rule. Resulting units are new stock. They start in quarantine until they are released, so a just-irradiated or just-pooled unit is not immediately available for issue. Release them from Inventory or the unit chart when the modification has been checked. Scan the unit number printed on the new label. The scan has to match that result. A blank or different number is a hard stop. A quarantine release that is not a modification result does not ask for this scan.

Thaw of a plasma product that your facility marked for auto-thaw can also happen at allocation. Those units expire 24 hours from the thaw time. The product row shows the new expiration. Do not issue on the old frozen expiration.

Pooling, irradiation, and similar changes may apply an expiration-modification code. The new date is calculated. If the date looks wrong, stop and check the modification rule and the expiration code with a supervisor. Do not overwrite the expiration by hand.

Supply lot

You can record a supply description and lot with the modification. Both are optional. They are stored on the modification, written to the audit record, and shown on modification history. Facility policy can require the lot. That policy ships off.

> Facility policy. When require supply lot is on, a blank lot is a hard stop and the units are not modified.

Batch

For irradiate, thaw, volume reduction, leukoreduction, and wash, you can apply the selected rule and the same reason to the unit on the form plus the unit numbers you list. If any unit is ineligible, none of them are modified.

Orders that asked for the modification

A product order can request irradiate, wash, thaw, volume reduce, or leukoreduce. Completing that action here satisfies the order and can queue the action charge. Complete the action only when the physical modification is done.

> Warning. Resulting units are quarantined on purpose. Issuing the source unit after it has been modified, or issuing the child unit before release, is blocked by unit status.

> Hard stop. A modification result stays in quarantine when the scanned unit number is blank or does not match the result.

Reagent QC

Record reagent lots, daily QC, and a review by a different person. The worklist warns. Patient resulting stays open unless facility policy is turned on.

Reagent QC is the bench record for reagents you use in testing. Add the reagent, receive a lot, open it when you start using it, and close it when it is finished. A lot stores the received, opened, and closed times and the expiration.

Each day, record the expected result, the observed result, and whether the row passed or failed. The person signed in is the performer. Review is a separate sign-in. The same person cannot review their own row.

> Hard stop. Review is blocked when the reviewer is the person who performed the QC.

The worklist shows lots with no QC for the current day, rows that failed, and rows that have not been reviewed. Closing a lot takes it off the overdue list.

> Facility policy. Patient resulting stays available while this worklist has rows. An administrator can turn on **Block results when reagent QC is open**. That policy ships off. It is not a CLIA hold.

This screen does not decide that a reagent is acceptable for a patient. It records what the bench did and who reviewed it.

Stock par

Compare available units with a minimum count by product and optional ABO. The board does not order blood.

Stock par shows how many available units you have against a minimum you set. The minimum is for one product. You can limit it to one ABO group, or leave ABO as Any so the count includes every available unit of that product.

On hand is the available count. Short is how many units you are under the minimum. A shortfall of zero means the available count meets the minimum.

An administrator with configuration edit permission saves the minimum. Saving again for the same product and ABO updates that row.

The board does not place an order with a blood center. It does not change unit status.

Compatibility and issue

Compatibility testing

Record a serologic or electronic crossmatch and reserve the unit to the patient.

Compatibility Testing is the desk for crossmatch and reservation when you are not already on the patient's Products tab. Identify the patient by MRN or name and the specimen by accession. The specimen must be current and must belong to that patient.

Record the crossmatch

Choose the unit and the method.

Serologic crossmatch asks for the crossmatch test and the reactions (the phases your test definition includes). Record what you saw. Incompatible is a result you can record. It does not become compatible because the patient is urgent.

Electronic crossmatch is available only when all of the following are true:

- Facility policy has electronic crossmatch turned on. It is off until an administrator enables it.
- The patient has a current ABO and Rh and a negative antibody screen that meets the visit, specimen, and test minimums in Crossmatch Settings.
- The patient has no antibody history of any kind. A deactivated antibody and an antibody marked currently undetectable still count.
- No antibody-identification workup is open.

If you opened the screen and someone else posted an antibody before you saved, the electronic crossmatch is refused. Record a serologic crossmatch instead.

> Hard stop. Electronic crossmatch is refused when policy disables it, when any antibody history exists, or when an identification workup is open.

Reserve the unit

After a compatible crossmatch, allocate the unit to the patient's product order. Allocation warns when the unit is not antigen-negative for a requirement the patient has, and when a red cell has not been retyped. Those warnings are there so you pick a different unit before issue, not so you can click past them without reading them.

An emergency or massive-transfusion issue that still needs a retrospective crossmatch can open this screen with the patient and unit already filled in. Record the crossmatch against the specimen you actually tested. The retrospective queue on the issue desk clears only when that crossmatch is filed.

Crossmatch from the chart uses the same rules in a dialog. See Products on the chart.

Issue and transfuse

Issue reserved units, release blood in an emergency, acknowledge ward receipt, document transfusion, and take units back.



Issue and Transfuse is the desk for blood that is leaving the bank and for blood that is coming back. One board is open at a time: Ready to issue, Second ABO/Rh, Retrospective, In transit, Issued units, Issue, Ward receipt, Return, and Document transfusion. The issue gate runs on every routine issue. It checks identity, the specimen, ABO and Rh, allocation, crossmatch, special requirements, appearance, autologous or directed restrictions, and whether the chart you are using was merged away. Storage-location rules and remote electronic issue are part of the same gate.

Ready to issue

Ready to issue opens first. Reserved units are listed so you do not retype the patient and the unit. The row tells you when something is wrong before you start:

- A red cell that still needs a second concordant ABO or Rh is not offered for routine issue.
- A missing or expired specimen is not offered for routine or emergency issue.
- An outdated unit is not offered.
- A unit that still needs a compatible crossmatch is not offered for routine issue.
- An open antibody workup is marked and can still be issued.
- A unit inside the near-expiry window is marked and can still be issued.
- When electronic crossmatch is not eligible, the row shows the reason. Serologic issue is unchanged.

Issue on the row switches to the Issue board with the unit, MRN, and date of birth filled in. A link that already names the patient or the unit opens that same board. Complete a second identifier, appearance, location, and who is taking the unit. Print the compatibility tag when the issue requires a P-tag, then scan the P-tag id before you issue. Facility policy decides whether the tag is required.

> Hard stop. Two independent identity tokens are required. Confirming a single checkbox does not satisfy the check. A hard stop cannot be signed away.

> Warning. A warning (near expiry, an antigen-negative gap, a historical ABO discrepancy that is overridable, or a complex crossmatch that was not done) asks for a reason, the authorizer, and an electronic signature. The signature is re-authentication. The override is stored with the issue. It does not change the result that caused it.

Emergency release and massive transfusion

Emergency and MTP are a different issue type. They are not an instruction to ignore every rule. The missing crossmatch becomes a warning only inside this issue type, and only for someone who has the emergency-release permission. Technologists who can issue routine units cannot release uncrossmatched blood unless that permission was granted.

You still need a current specimen for the patient, an in-date unit, acceptable appearance, and two identity tokens. You record the reason and sign. The issue is flagged for retrospective crossmatch. It stays on the retrospective queue until someone records the crossmatch, by default within the hours set in facility policy (the shipped default is 24). **Record crossmatch** opens Compatibility with the issue filled in.

In transit and ward receipt

In transit lists units that left in a cooler or with a courier until the receiving location acknowledges custody. **Receive** on the row opens Ward receipt. Receipt repeats the product, expiration, type, specimen, crossmatch, who it was issued to, the cooler, and the destination so the person receiving can compare them with the unit in hand. Transfusion documentation waits until receipt is recorded when that is the path the unit took.

Document the transfusion

Issued units lists outstanding issues so you can document or return them without looking up an internal issue number. **Document** opens Document transfusion and records the identifiers checked at the bedside, the start, and the disposition your procedure captures. Dual identification is required here the same way it is required at issue.

Final disposition

Final disposition lists issued units that are past the facility window and still have no transfusion document, return, or discard. The window ships at 24 hours. Change it in Facility Policy if your procedure uses another operational interval. The board is a warning. It does not stop you from issuing another unit, documenting the transfusion, returning the unit, or discarding it.

> Warning. A unit on this board still needs a final fate. The list does not change the unit.

Return

Return on an issued row opens the Return board for a unit that came back unused. Record temperature, seal, and appearance. A unit that fails those checks does not go back to available stock. The issued-units board shows whether the storage device is governed. A governed device cannot return to available stock after the facility window, which ships at 30 minutes. Saying the time out was acceptable does not change that. An exempt device does not fail the window. Reissue follows the return rules on the product. Do not document a transfusion for a unit you are returning, and do not return a unit that was transfused.

What the gate will not do

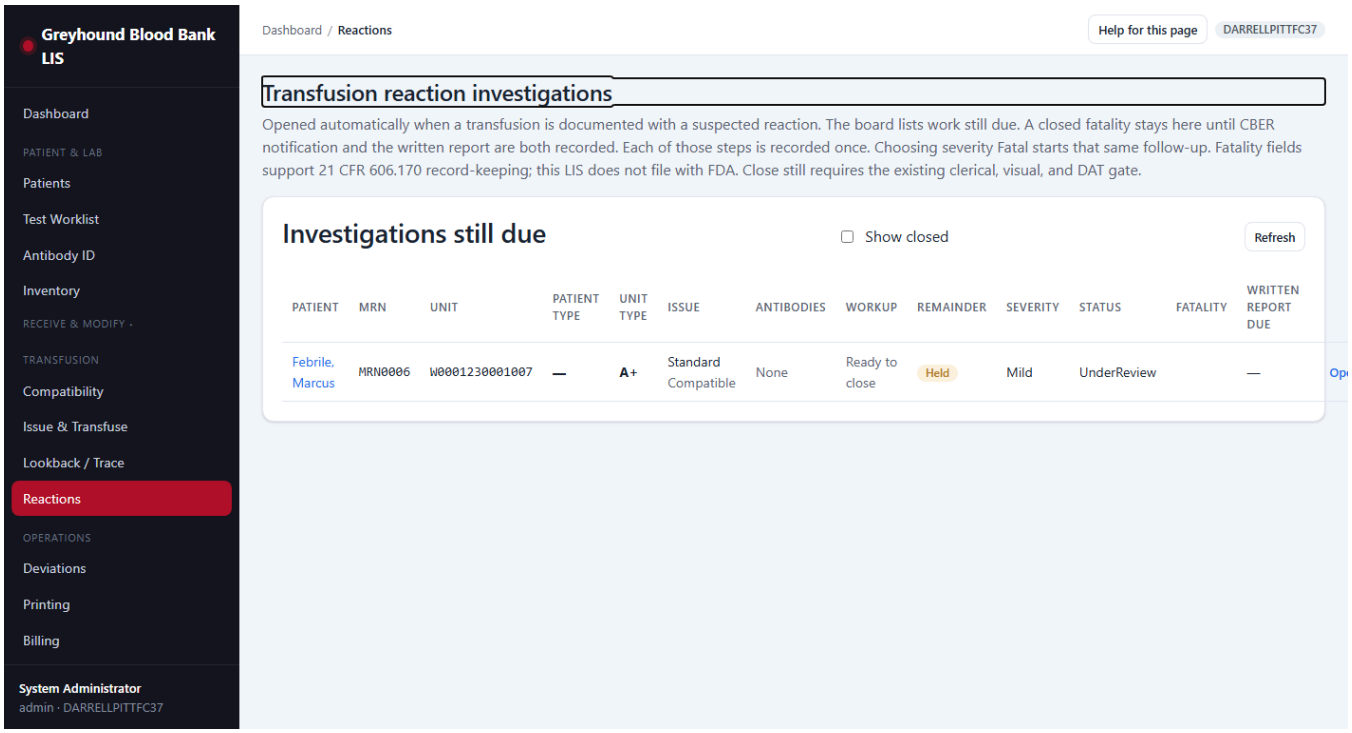
The gate will not issue the losing chart of a merged patient, a unit allocated to someone else, an autologous or directed unit to the wrong patient, a unit that fails appearance, or a unit that does not meet an enforced special requirement. Fix the record or choose another unit. See Safety checks for the list you will see most often.

Quality and operations

Transfusion reactions

Investigate a suspected reaction, quarantine remaining units, and close the investigation only when the checks

are done.



Reactions is the investigation board. You need permission to investigate reactions. Open an existing investigation or start one from the transfusion it belongs to.

What you record

Record the clerical check, the visual check of the unit and the specimen, and the DAT, along with the symptoms and the timeline the ward reported. The investigation stays open while those items are incomplete. Closing is a separate step with its own gate and a signature when the close action requires one.

Remaining product from the implicated unit is quarantined so it cannot be issued to someone else while you are investigating. Do not release that quarantine to "put the unit back" until the investigation says the unit can be released. Discard, if that is the outcome, uses the inventory discard permission and a reason.

Fatality and lookback

A reaction that your procedure treats as a fatality or as a product-safety signal stays open for the follow-up your medical director requires. The screen keeps that follow-up visible. It does not file a regulatory report for you. Filing outside this system is a facility procedure.

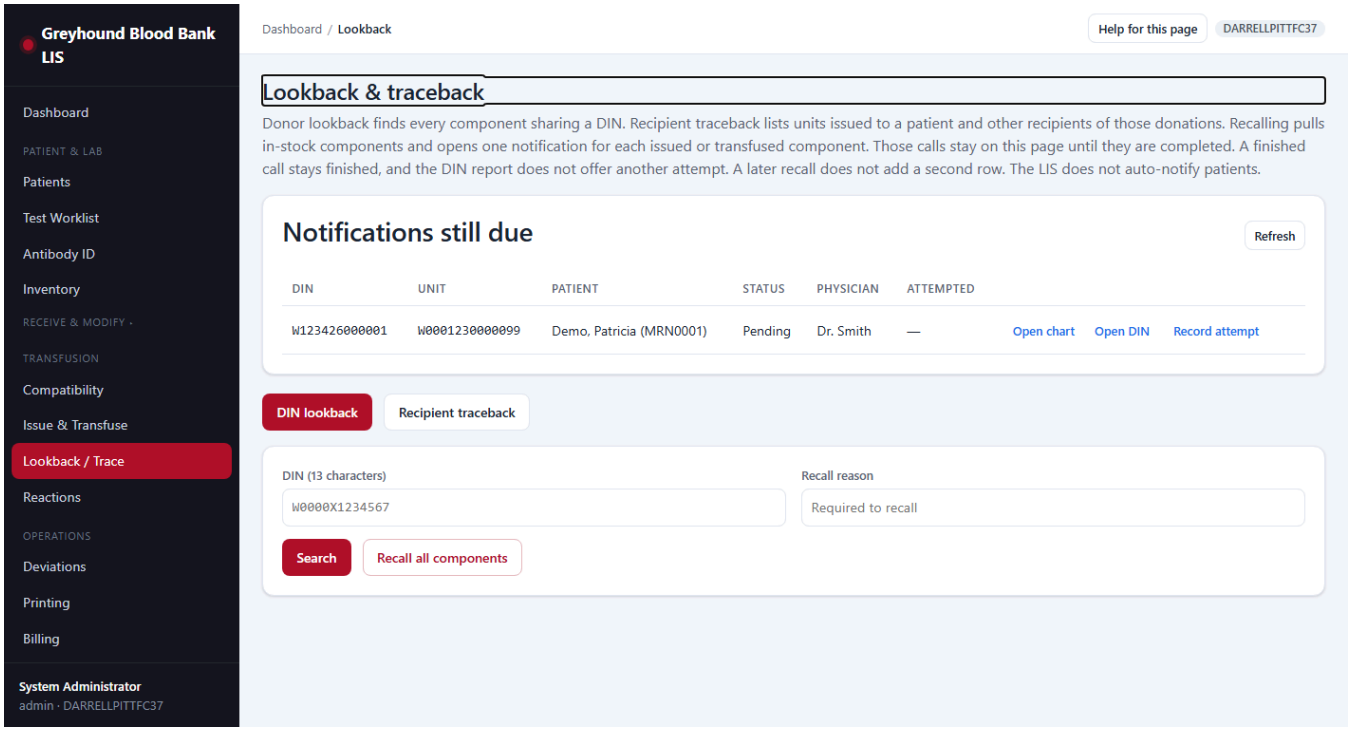
If the investigation implicates a donor unit, start or continue the Lookback from the donation identification number. The reaction and the lookback are separate records that reference the same unit.

> Hard stop. Closing is refused while required clerical, visual, or DAT items are unfinished. Do not close

an investigation to clear the board.

Lookback and traceback

Start from a donor unit to find recipients, or from a patient to find the units they received, and record notifications.



Lookback / Trace covers two directions.

Donor lookback starts with the donation identification number. The screen lists recipients who received a product from that donation and the units involved. Use it when a supplier notifies you about a donor.

Recipient traceback starts with the patient MRN. It lists the units that patient received so you can see which donations are involved.

Notifications

Record each notification you make: who was called, when, and the outcome. The record is the evidence that the call happened. The software does not place the call and does not send the letter. A recall that needs a unit pulled from inventory is still an inventory action: quarantine or discard the unit, with the lookback as the reason.

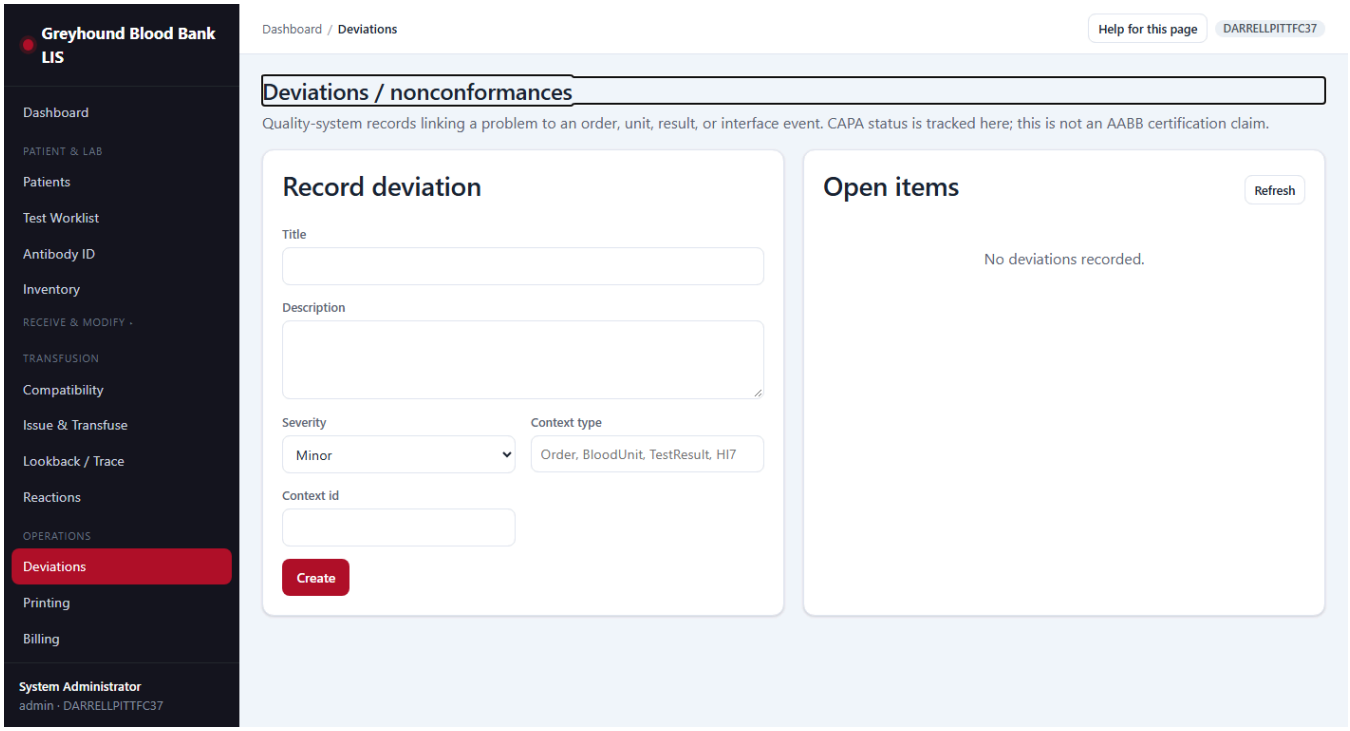
You need lookback permission. Other staff can see the clinical chart but cannot open this desk.

Facility policy. How fast you notify, and which events require a lookback, are your procedure and the

supplier's instruction. The screen stores what you did. It does not decide that a notification was required.

Deviations

Record a nonconformance, link it to the patient, unit, or order it affects, and track it to closure.



Deviations is the quality record for something that did not follow procedure: a wrong unit selected and caught, a labeling error, a temperature excursion, a missed retrospective crossmatch, an interface result filed on the wrong patient. You need permission to manage deviations.

Write the record

Describe what happened, when it was discovered, and what the immediate risk is. Link the patient, specimen, unit, order, or issue that is involved so the next person can open the chart without searching. The link is a reference. It does not by itself quarantine a unit or invalidate a result. Do those actions in Inventory or on the result, and cite the deviation.

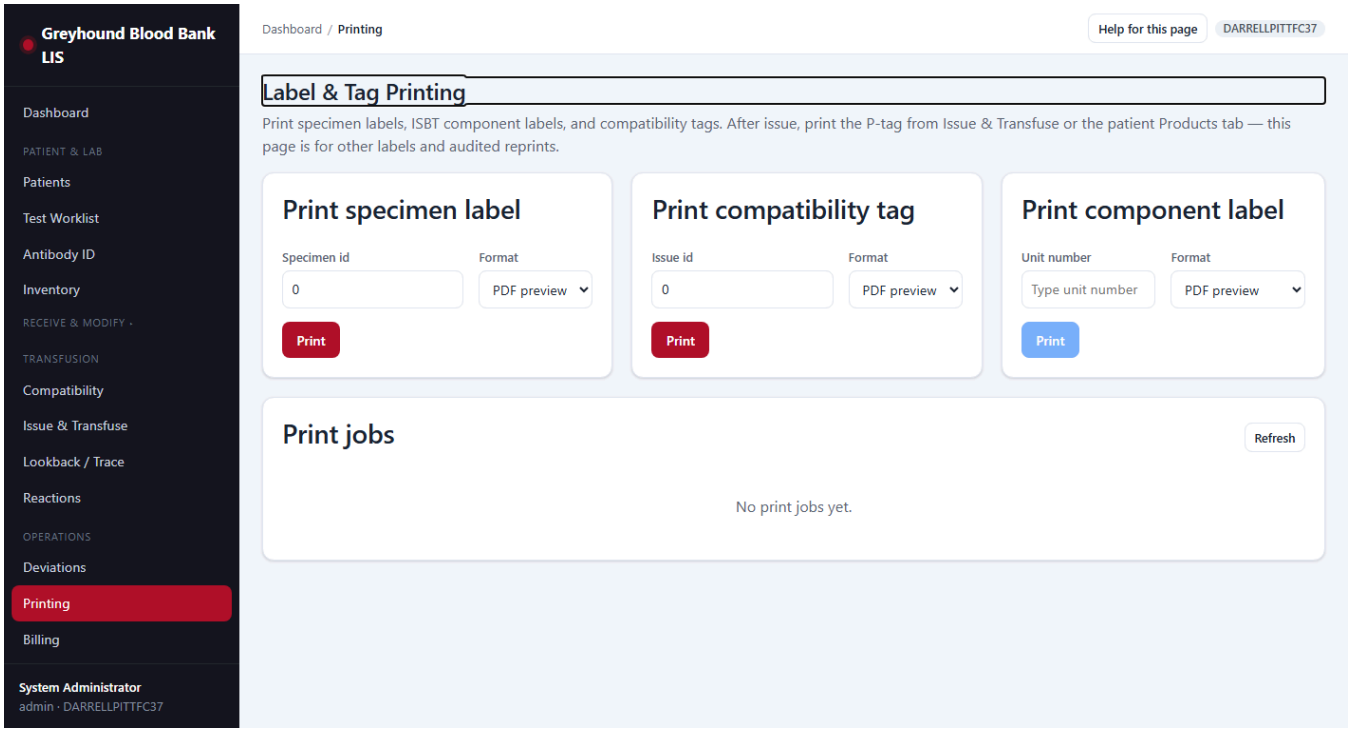
Status moves from open toward closure as the corrective action is done. Closure requires the outcome, not a blank completion. The record stays in the audit trail. You cannot delete a deviation to remove an uncomfortable event.

Use a deviation in addition to an issue override, not instead of it. The override explains why the gate allowed the issue. The deviation explains the process failure if there was one.

Interfaces and reports

Printing

Print specimen labels, component labels, and compatibility tags, and reprint only with a reason.



Printing produces specimen labels, component labels, and compatibility tags (P-tags). You can also open it from a specimen, a unit, or an issue, with that record already selected.

What you are printing

Confirm the identifiers on the preview before you send the job. The preview is a PDF. The printer may be a PDF queue or a Zebra printer that takes ZPL. Which printers exist, and which layout is active for the specimen label and the P-tag, is set in Administration. You pick the printer. You do not redesign the label here.

A compatibility tag printed at issue carries the patient, the unit, and the crossmatch information the layout includes. When facility policy requires the P-tag id before issue, scan that id on the issue form. Printing the tag and not scanning it does not satisfy the check.

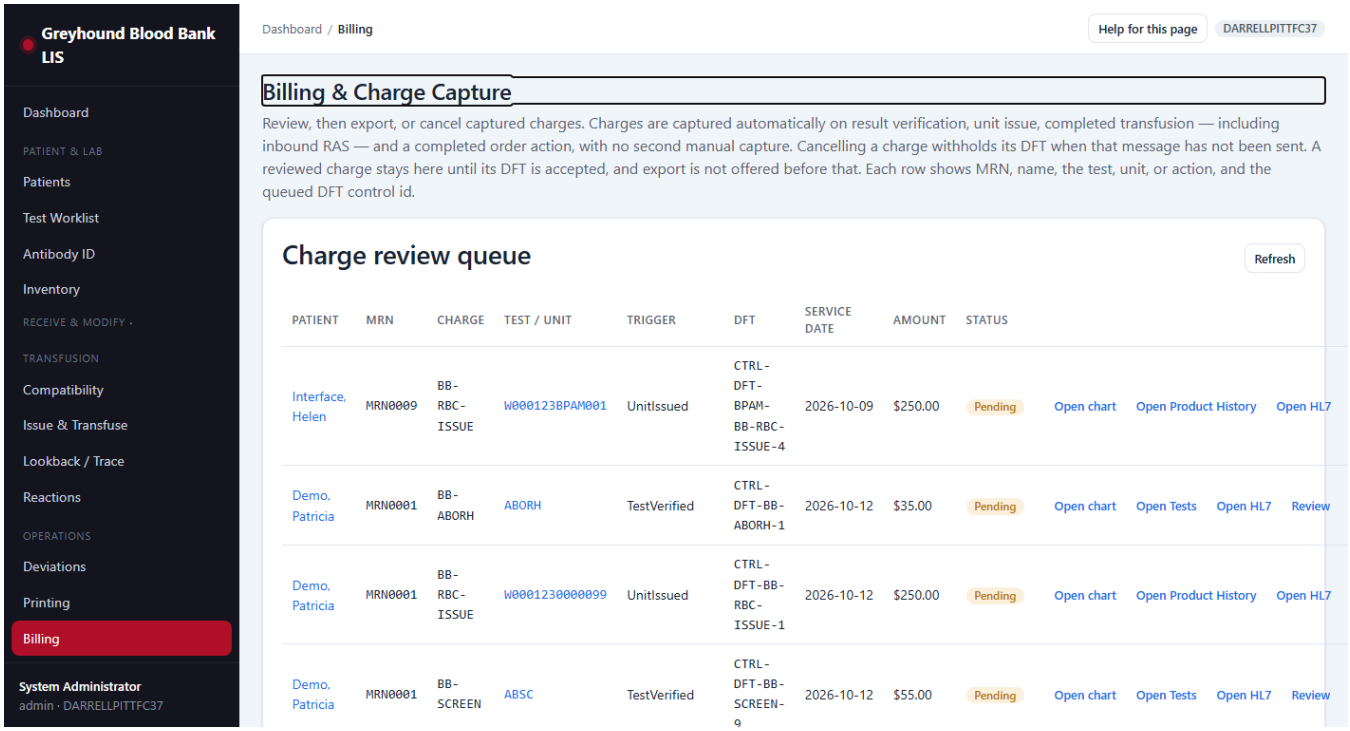
Reprint

Reprint requires a reason and the reprint permission. The reason is audited. Reprint when the label is damaged or lost. Do not reprint onto a different specimen or unit. If the identifiers on the label are wrong, fix the record first, then print again, or reject the specimen and start over.

Jobs stay in the queue on this screen so you can see what was sent and open the preview again.

Billing review

Review charges captured from verified tests, issued units, and completed actions, then export or cancel them.



Charges are captured when a result is verified, a unit is issued or transfused (as the charge rule says), or a modification action is completed. You do not key a charge for a successful clinical action. **Billing** is the review queue for charges that were captured and for the outbound DFT messages waiting to go to the billing system.

Review and export

Open the queue if you have billing review permission. Confirm the patient, the code, and the triggering event. Export sends the accepted batch. Export can stay blocked until the outbound DFT is accepted. If a row will not export, read the status rather than entering the charge in another system and exporting it here as well.

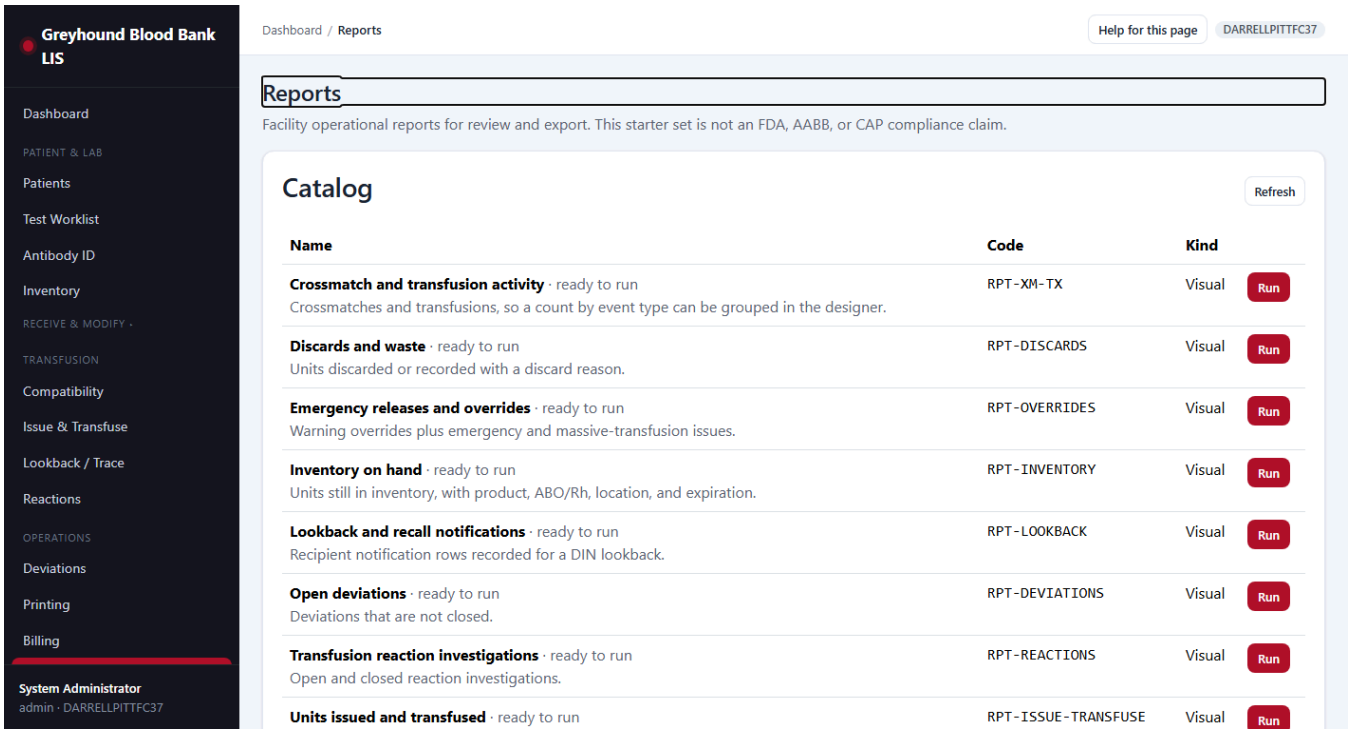
Cancel

Cancel requires a reason and the cancel permission. Cancel when the clinical event was reversed or charged in error, for example a result that was invalidated or an issue that was returned under your billing rules. Do not cancel a charge only because the dollar amount looks high. The amount comes from the charge master. An administrator changes the master. You document the reason for a cancellation.

A charge you cannot see means you lack review permission, not that the capture failed. Ask a billing analyst to open the queue.

Reports

Run a report from the catalog and export it to PDF or Excel.



Reports lists the reports you are allowed to run. Designing a report is a separate administrator screen. Running one does not change clinical data.

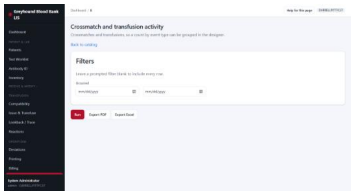
Open a report, enter the parameters it asks for (a date range is typical), and run it. The result is a table. Export PDF or Excel from the run screen when you need a file. Large results can be truncated at the export limit. The export says so when that happens. Narrow the dates rather than assuming the missing rows were blank.

Operational reports are facility reports. They are not a regulatory certificate. Read the title and the parameters you typed before you send the file to someone else.

If the report you need is not listed, you either lack report permission or nobody has published that definition. Report design is in the Administrator Guide.

Running a report

Enter parameters, read the rows, and export the run to PDF or Excel.



The run screen is one report. The title is the report name. Parameters are the filters the designer defined. A date range that is backwards or empty returns no rows. That is a parameter result, not a failure of the clinical record.

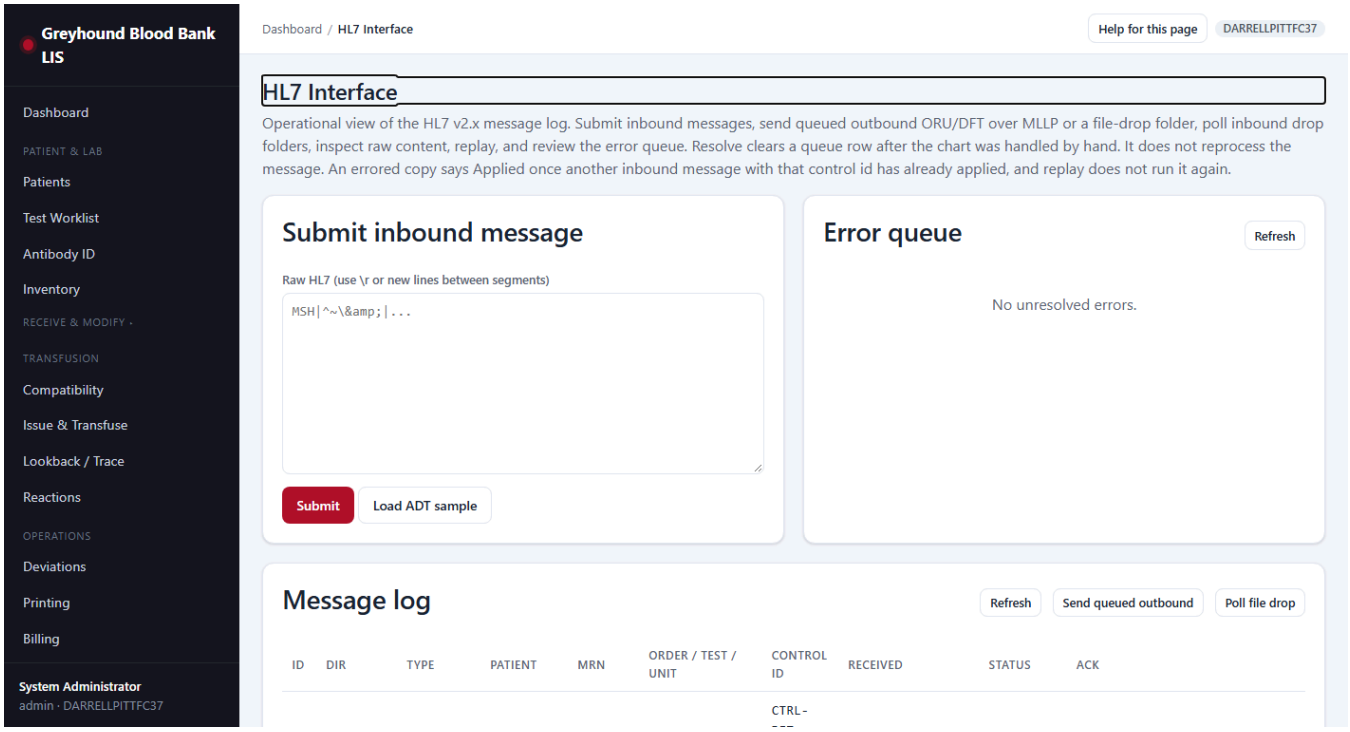
Run, then read the table before you export. Export PDF for a paginated copy and Excel when you need to sort or filter further. The PDF is marked as a facility operational report. It is not an accreditation certificate.

If the report was changed while you had it open, run it again. The file you export is the run you just executed, not a later edit to the design.

To change columns or the query, you need report-design permission and the designer. Running permission is not design permission.

HL7 interface console

Read the message log, open errors, and replay a failed message when you manage the interface.



The **HL7 Interface** console is the operational log. Admission, order, and result messages are listed with their status. You need permission to manage HL7 to open it. Configuring the connection, the field map, and the translation table is an administrator task. This screen is for seeing what arrived and what failed.

Errors

Open an error and read the reason. A rejected order or result did not update the chart. Replaying a message sends it through processing again. Replay when the failure was transient or when the map has been fixed. Do not replay a message that filed on the wrong patient until the map or the translation is corrected, or you will file it wrong again.

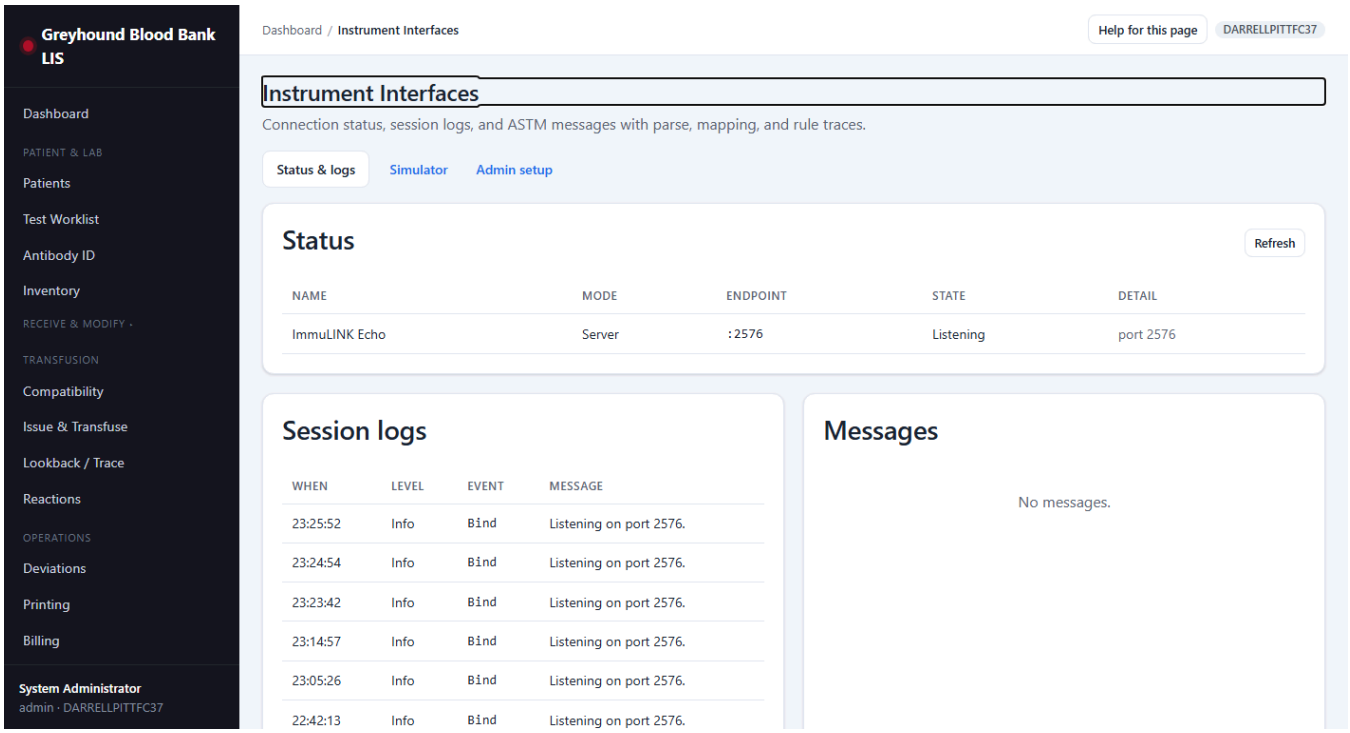
Outbound messages, including billing DFT and result messages your interfaces send, appear with their acknowledgement status. A message that has not been acknowledged is still pending. The downtime snapshot counts those so you can see the backlog after an outage.

What the bench should do

If an order you expected is missing, check this log before you place a duplicate. If the message failed, fix or replay it. If it succeeded, the order is on the chart and you may be searching the wrong MRN. Placing a second manual order for a message that already succeeded creates two orders for one request.

Instrument interfaces

See whether each instrument connection is receiving messages and open the log when a result did not file.



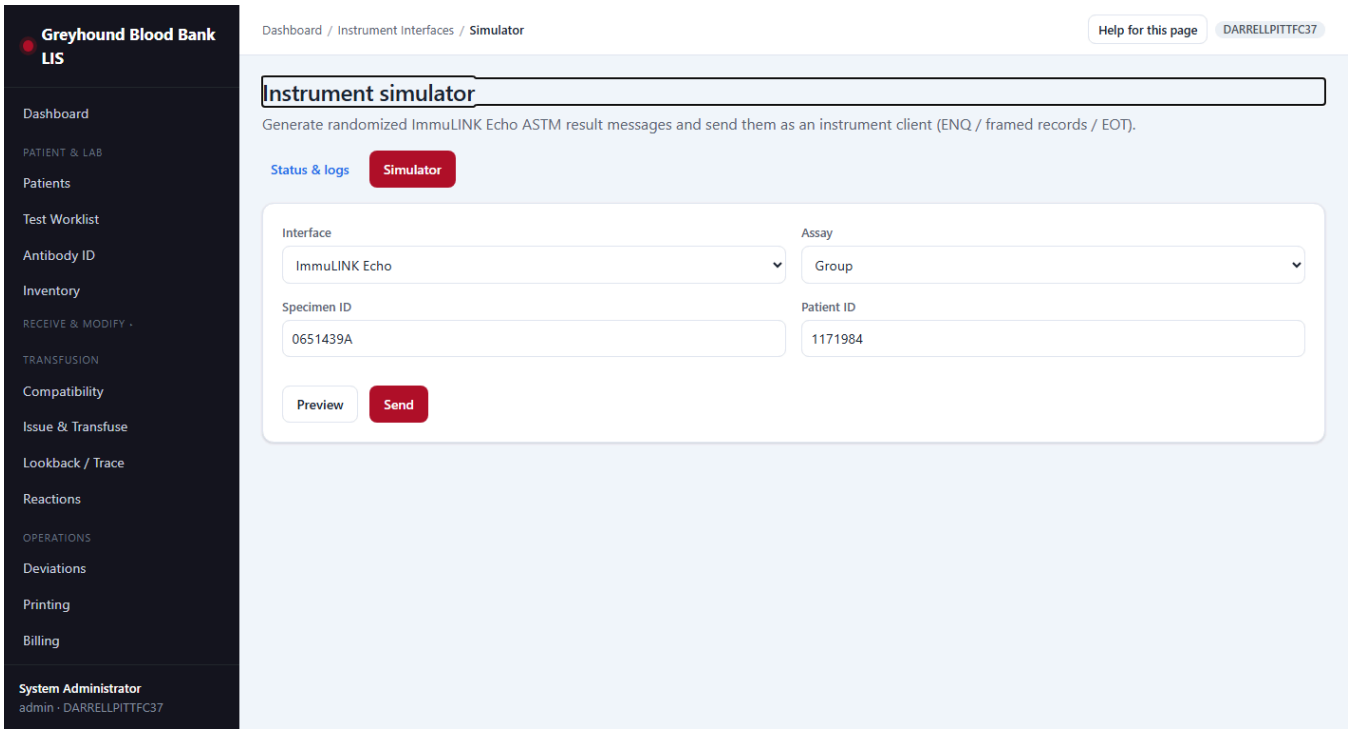
Instrument Interfaces shows the status of ASTM instrument connections and their recent messages. You need permission to manage instruments. The mapping of instrument tests onto LIS tests, and the if/then rules, are configured in Administration. The help panel on that administrator screen is the field reference. This page tells you whether messages are arriving.

When a result did not appear on the worklist, open the log for that instrument. A message that failed to map names the test code the instrument sent. Do not retype the result from the instrument printout and also leave the interface message to file later. You will have two results. Either replay after the map is fixed, or enter the result manually and resolve the interface message so it does not file again.

The simulator is a separate screen for sending a test message. Do not point it at a live instrument connection.

Instrument simulator

Send a test ASTM message to confirm a mapping. This is a testing tool, not a clinical result entry screen.



The **Instrument Simulator** sends an ASTM message you compose into the interface so you can see how it maps. Use it when an administrator has changed a test map and you need a dry run, or when you are investigating a message format.

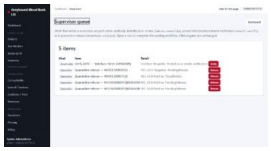
A message the simulator files is a real result if it matches a real specimen and the interface accepts it. Do not send a message that names a clinical accession unless you intend to create that result. Use a training accession on a training database.

This screen does not replace result entry. Clinical results are entered or verified on the worklist, including results that arrived from a real instrument.

Supervision

Supervisor queue

Review antibody identifications, verify results that are waiting, and release quarantined units that need a higher permission.



The **Supervisor queue** collects work that a technologist cannot finish alone:

- Antibody-identification workups waiting for review.
- Results waiting for verification when you are the verifier.
- Quarantine releases that require the release permission and, when policy says so, a second verifier.

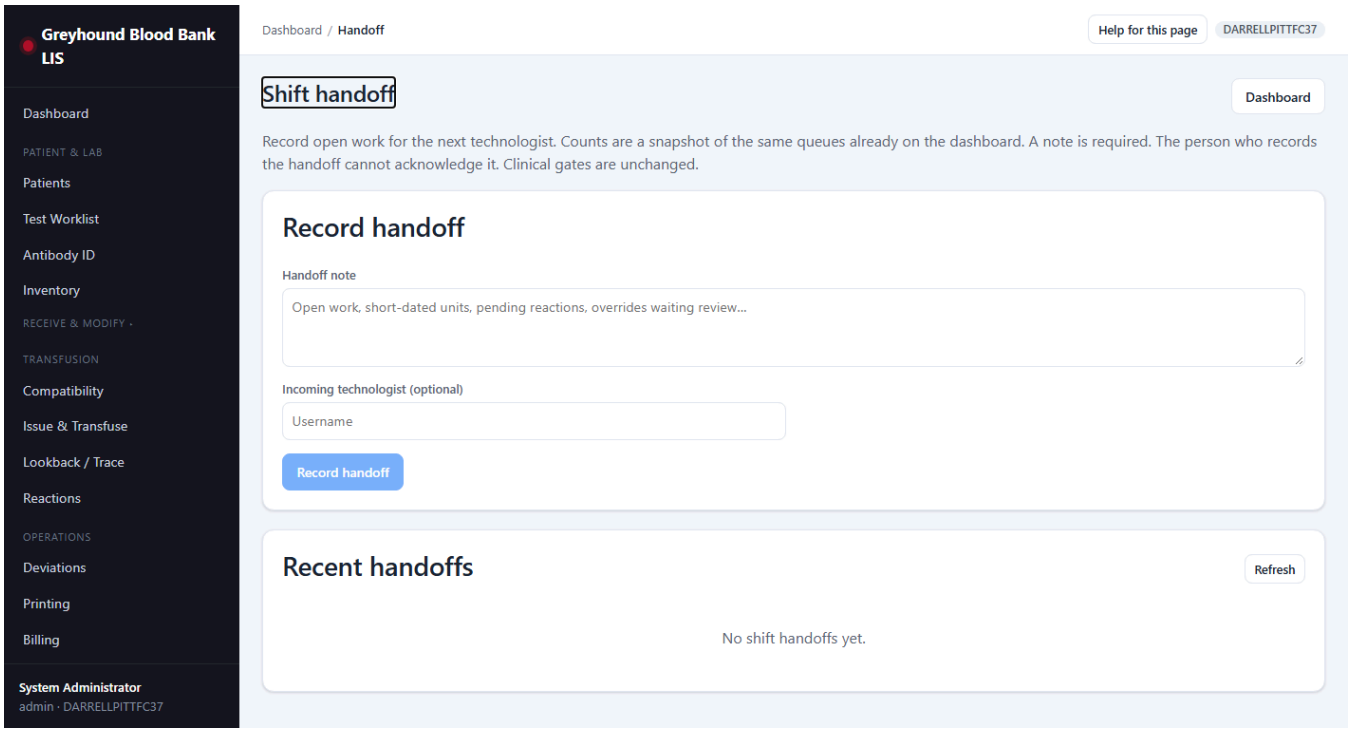
Open the row, read the underlying record, and complete the review on that record. The queue is a list. The decision is recorded on the workup, the result, or the unit, not as a checkmark that skips the screen.

You see the sections your permissions allow. A supervisor who cannot release inventory will not see release rows. If a technologist says an item is "in the queue" and you do not see it, confirm you are signed in as yourself and that the item was actually submitted.

> Warning. Releasing a unit from quarantine makes it selectable for patients. Read the quarantine reason before you release. A retype mismatch is not a routine release.

Shift handoff

Read the queue counts at shift change and write the note the next shift must see.



Shift Handoff shows the counts that matter at change of shift: work still open on the benches you can see, including pending results, open issues, and retrospective crossmatches that are due. You need audit read permission to open it.

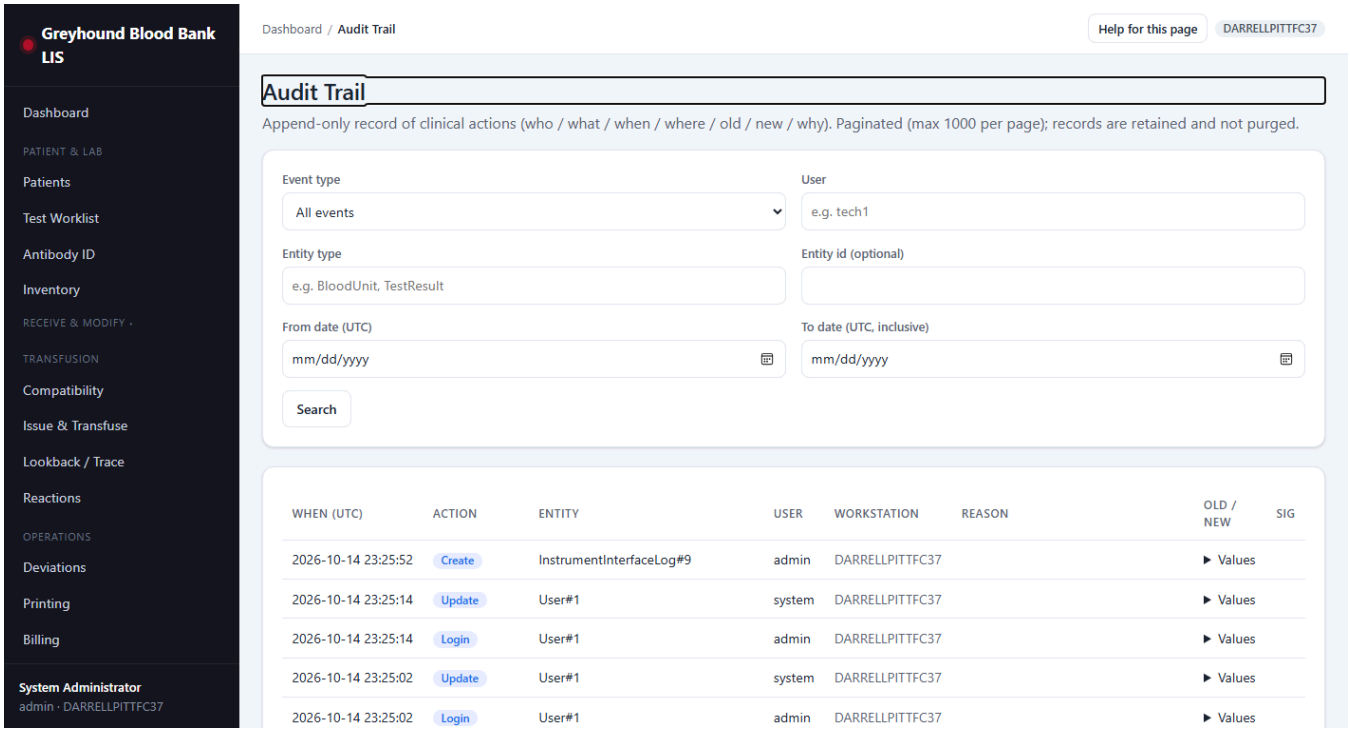
The counts are a snapshot, not a sign-out of each patient. Write the handoff note for anything the count cannot say: a unit in a cooler that has not been received, a patient whose specimen is unacceptable, an

interface that has been down, a retrospective crossmatch that is overdue. The note is required. An empty handoff is not a completed handoff.

The next shift should read the note before relying on the dashboard alone. The note does not assign tasks inside the software. It is the record of what you said you left.

Audit trail

Filter the append-only audit log by user, patient, unit, or time.



The **Audit Trail** records who did what, at which workstation, and when. It includes clinical actions and the reasons captured with them: overrides, corrections, reprints, discards, and configuration changes. You need audit read permission. You do not need to be an administrator to read it, if that permission was granted to your role.

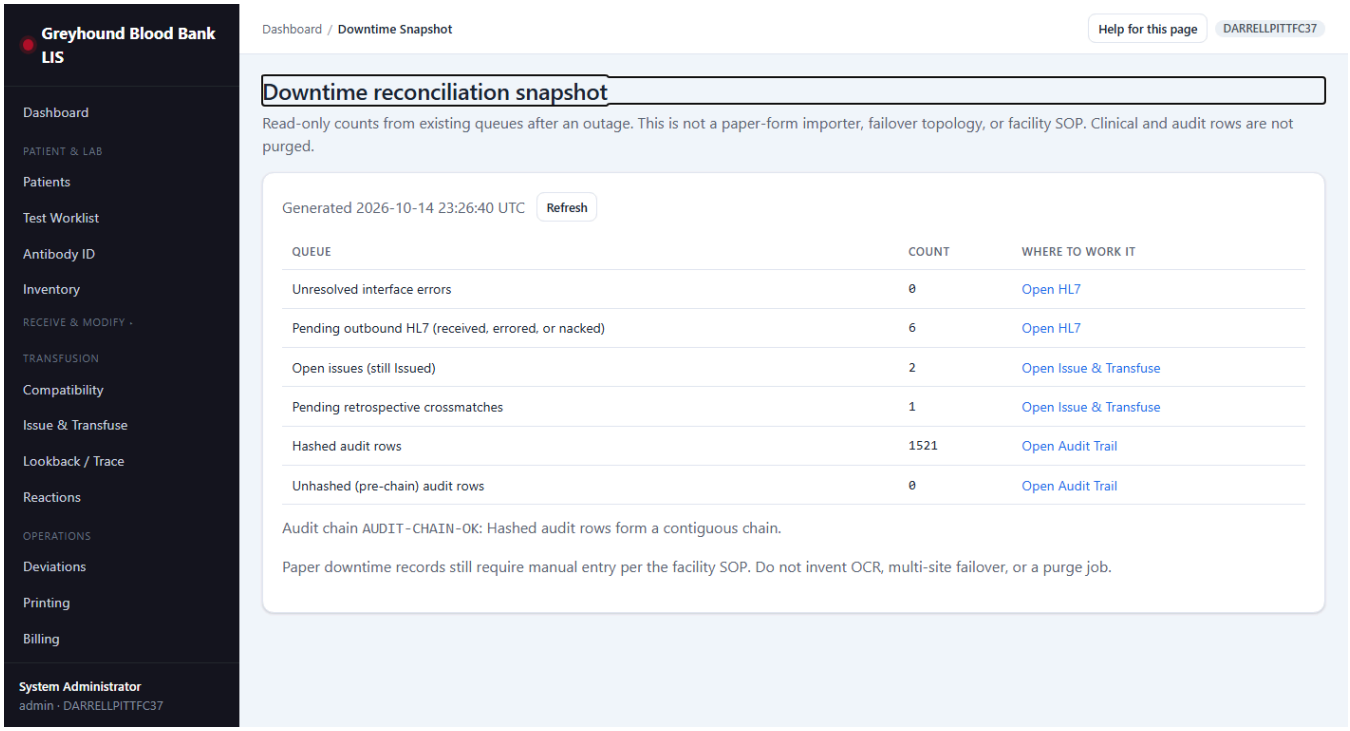
Filter by time, user, patient, or unit. Open the row to see the detail. The log is append-only. There is no delete and no edit. A new hash chain covers rows written after the chain was turned on. Older rows were not rewritten. If a chain check fails, treat that as a quality event and do not "fix" it by changing data.

Use the audit trail to answer a specific question. It is not a report designer. For a table you need to export on a schedule, use Reports.

Configuration changes also appear in **Configuration History** under Administration, which compares versions. The audit trail is the broader record, including bench actions.

Downtime snapshot

After the system returns, review interface errors, open issues, and retrospective crossmatches that still need to be entered from paper.



Downtime Snapshot is the recovery checklist inside the application. It counts unresolved interface errors, outbound HL7 that is still pending, open issues, retrospective crossmatches that are due, and the status of the audit chain. You need audit read permission. The screen does not import paper forms and it does not scan them.

While the system is down

Downtime documentation is your facility's paper procedure. Capture the identifiers you will need to enter later: patient MRN and name, specimen accession and collection time, ABO and Rh, antibody screen, antibodies and special requirements, unit number or donation identification number, product, ABO and Rh, expiration, inspection, crossmatch or the emergency-release acknowledgment, and any transfusion or reaction. The software does not store those paper rows for you.

After the system returns

1. Confirm you are on the recovered database, not a partial copy.
2. Replay interface messages that failed, and watch for duplicates. A message that already filed must not be entered by hand as well.
3. Enter the paper record into the chart, the specimen, the result, and the issue screens. Use the same identity checks you use for any issue. The first electronic issue after downtime is not a shortcut.
4. Complete retrospective crossmatches for emergency issues that were documented on paper.
5. Review overrides and interface errors before you call the recovery finished.

> Facility policy. The paper forms, the decision to transfuse during downtime, and the regulatory notice are your procedure. This screen only shows what is still open in the database.

Safety

Safety checks

How hard stops differ from warnings, what the issue gate checks, and when the system asks for an electronic signature.

The bench does not decide compatibility by itself. The application runs the safety rules and returns a result: pass, warning, or hard stop. The alert on the screen is the message you act on. This topic is the map of the checks you will meet most often.

Hard stop and warning

A **hard stop** ends the action. There is no reason field that lets you continue. Correct the record or choose a different patient, specimen, or unit.

A **warning** lets a person with the override permission continue after a reason and, for issue warnings, an electronic signature. The warning and the override are both stored. Overriding does not change the antibody, the crossmatch, or the blood type.

> Hard stop. Do not use another user's session to get past a check you cannot pass. The audit trail will show their name, and the check exists because the action is unsafe.

Issue gate

Before a unit leaves the bank, the gate checks:

Check	What it means
Identity	Two independent tokens, such as MRN and date of birth, at issue and again when transfusion is documented.
Specimen	A current specimen for this patient, not expired and not rejected.
ABO and Rh	The patient has a type, the unit has a type, and they are compatible for the component.
Unit status	The unit is in a status that can be issued, is not expired, and is the unit reserved to this patient.
Crossmatch	A current compatible crossmatch, unless this is an emergency or MTP issue.
Special requirements	Active requirements such as irradiation or antigen-negative are met.

Appearance	The unit passed visual inspection at issue.
Autologous and directed	The unit is issued only to the intended patient.
Merged chart	The MRN is the surviving chart, not the one that was merged away.

Emergency and MTP do not skip identity, specimen, expiration, or appearance. They change the missing crossmatch from a hard stop into a warning, and only for a user with emergency-release permission. The issue is then due for a retrospective crossmatch.

Electronic crossmatch

Electronic crossmatch is off until Facility Policy turns it on. Even then it is refused when the patient has any antibody history, including deactivated rows, or an open identification workup. It is also refused when the patient does not meet the minimum negative antibody screens configured in Crossmatch Settings. Serologic crossmatch remains available.

Signatures

The system asks you to sign by entering your password again. The signature is stored with the action. You will be asked when you:

- Override an issue warning, including emergency release without a crossmatch.
- Correct a verified result.
- Override ABO or Rh history.
- Close a reaction investigation when the close action requires a signature.

Discard, return, reprint, and product modification ask for a reason. A signature is added when that action's policy requires one. Reprint specifically requires the reprint permission and a reason.

Antibodies

Antibody history is append-only. Deactivate means "not currently detectable," not "never identified." Any historical row blocks electronic crossmatch. An open identification workup blocks chart edits to antibodies and blocks electronic crossmatch. Completing the workup posts the identified specificities. Assist on the workup never identifies the antibody for you.

Appendix

Training data

Demo accounts and example patients that exist only on a database seeded for training or development.

A development or training database can be seeded with demo accounts and example patients. A clinical database should not keep the demo password. If you can sign in as **admin** with the password **demo**, you are not on a production system.

Demo accounts

The password for each of these accounts, on a seeded training database, is **demo**.

Username	Typical role	Use it to see
admin	Administrator	Every screen, including Administration
supervisor	Supervisor	Emergency release, overrides, merge, discard, reaction and lookback work
tech1	Technologist	Everyday bench work without emergency release or user administration
viewer	Read only	Audit and reports, without entering results
analyst	Interface analyst	HL7 and instrument consoles
invmgr	Inventory manager	Inventory operations and product setup
biller	Billing analyst	The billing queue
auditor	Auditor	Audit, reports, and configuration history

Signing in as one of these accounts shows the sidebar that role is allowed to see. The permission list on the dashboard is the exact set.

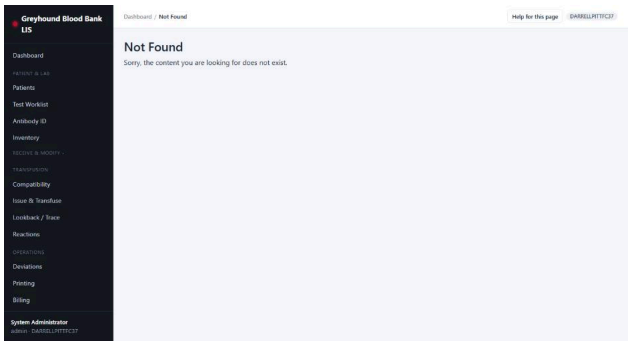
Example patients

Seeded charts are fictitious. They are useful for walking through this guide. MRN0001 (Patricia Demo) is a general chart. Other seeded MRNs illustrate a newborn, an Rh-negative patient, antigen-negative selection, an emergency release with a retrospective crossmatch due, a reaction investigation, autologous and directed units, and an electronic-crossmatch example. Electronic crossmatch on that example still requires Facility Policy to enable it.

Do not document training steps against a real MRN. Do not copy a screenshot from this guide into a patient chart.

When something goes wrong

What to do when a page is missing, an action is blocked, or the application shows an error.



The page is not in the sidebar

The sidebar hides screens your session cannot open. Check **Your access** on the dashboard. If the permission is missing, an administrator has to grant the role. Signing in again is required after a role change.

A web address that does not match a screen shows **not found**. Use the sidebar or Help rather than editing the address. The not-found page is not a clinical record.

The action is blocked

Read the alert. A hard stop names the rule that failed. Fix that condition. A warning names the override. If you do not have the override permission, the warning is as far as you can go. Find a supervisor rather than trying the action on a different screen. The Products tab, Compatibility, and Issue and Transfuse all run the same gate.

If Help for this page opens Safety checks, you are on a screen whose block is one of the issue or crossmatch rules.

The application shows an error

An unhandled error offers **Reload**. Reload once. If the error returns, note the screen you were on and the time, and tell the person who supports the system. Do not re-enter the same save repeatedly. You may create a duplicate order, result, or issue if the first save succeeded and only the confirmation failed.

If you were signed out, the session expired or was revoked. Sign in again. Unsaved fields on the form are not kept.

Figures in this guide

Screenshots in the PDF and in Help are from the training database. Your facility's catalog names, printers, and policies will differ. The steps are the same. The labels on a few administrator screens follow your configuration.